WELLNESS STUDIO OS

TECHNICAL SPECIFICATION · DOCUMENT 3 OF 6

Module Specification

Implementation-Ready Functional Blueprint

DOCUMENT	Document 3 of 6 — Module Specification
VERSION	1.0
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Document Control

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Classification	Confidential — Engineering, QA, AI and ERP Teams
Scope	10 sections: module inventory; B2B and B2C module specifications; screen inventory; cross-module workflows; AI modules and agents; automation workflows; dependencies and integration contracts; MVP implementation scope.

Priority Notation

Priorities used throughout this document:

P0	Mandatory 5-Day MVP
P1	Phase 1
P2	Phase 2
P3	Phase 3
F	Future

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SECTION 1

Complete Module Inventory

A B2B Modules (Studio Management Portal)

#	Module	Purpose	Priority	Key Dependencies
1	Dashboard	KPI overview, quick actions, renewal pipeline visibility for daily operations	P0	Members, Memberships, Bookings, Attendance, Payments, Reports
2	CRM	Lead capture, pipeline management, source attribution, lead-to-member conversion	P0	Members, Notifications, Events, Payments
3	Leads	Sub-module of CRM; individual lead records, follow-up notes, status, conversion flow	P0	CRM, Members, Notifications, WhatsApp
4	Members	Single-source member identity and lifecycle — create, read, update, search, dedup	P0	Memberships, Bookings, Payments, Attendance, CRM
5	Memberships	Studio plan definitions, member-plan assignment, approval workflow, expiry, renewal pipeline	P0	Members, Payments, Notifications, Classes, Settings
6	Classes	Class type configuration, session scheduling, trainer assignment, capacity, recurring schedules	P0	Bookings, Attendance, Staff Management, Settings
7	Bookings	Eligibility enforcement, atomic capacity management, waitlist, cancellation cutoff, book-on-behalf	P0	Members, Memberships, Classes, Payments, Notifications
8	Attendance	QR check-in, trainer marking, audited admin correction, attendance history	P0	Bookings, Members, Classes, Staff Management
9	Events	Workshops, studio events, public events — capacity, audience type, paid/free, registration	P0	Members, Payments, Notifications, Attendance, Settings
10	Payments	Razorpay online gateway, manual payment recording, status tracking, reconciliation	P0	Members, Memberships, Events, Notifications
11	Reports	Operational and financial insight — members, revenue, occupancy, attendance, renewals, leads	P1	All modules (read-only aggregates via reporting service)
12	Staff Management	Staff accounts, role assignment, granular permission grants, trainer assignment, deactivation	P0	All modules (role enforcement across platform)
13	Notifications	WhatsApp workflow templates, delivery status tracking, mock/live mode, failure visibility	P0	All event-producing modules, n8n, WhatsApp Cloud API

#	Module	Purpose	Priority	Key Dependencies
14	Settings	Configuration Center — plans, class types, branding, templates, timings, feature flags	P0	All configurable modules across platform
15	Tenant Management	Multi-tenant SaaS isolation, Tenant→Studio→Branch hierarchy, tier entitlement flags	P0	All modules (tenant scope enforcement)
16	AI Assistant (B2B)	Natural-language operational queries and Create Member action with confirmation	P1	Members, Memberships, Attendance, Reports, AI Platform

B B2C Modules (Member Portal)

#	Module	Purpose	Priority	Key Dependencies
1	Dashboard	Member personal view — membership status, upcoming bookings, renewal call-to-action	P0	Memberships, Bookings, Attendance
2	Membership	Browse studio plans, purchase, renew, view validity, benefits and history	P0	Payments, Settings, Notifications
3	Bookings	Live schedule availability, book, cancel within window, join waitlist, history	P0	Classes, Memberships, Payments, Notifications
4	Attendance	QR self-check-in, personal attendance history view	P0	Bookings, Classes, QR Generator
5	Events	Browse public and member events, register, pay if required, view own registrations	P0	Payments, Notifications, Settings
6	Payments	Online checkout, own payment history, transaction status	P0	Gateway (Razorpay), Memberships, Events
7	Profile	Own details management, emergency contact, phone OTP and Google login, account linking	P0	Auth, Notifications
8	Notifications	Receive WhatsApp confirmations and reminders; in-app notification centre (Phase 2)	P1	n8n, WhatsApp Cloud API, Settings
9	Recommendations	AI class suggestions based on stated goal, level, eligibility, and attendance history	P1	Classes, Memberships, AI Platform, Attendance

C AI Modules

Phase 1 MVP Assistants (Narrow Scope)

#	Module	Purpose	Priority	Key Dependencies
1	AI Admin Assistant	Operational queries (expiring memberships, low attendance) + Create Member with confirmation	P1	Members, Memberships, Attendance, Reports, LLM Gateway, AI Audit

#	Module	Purpose	Priority	Key Dependencies
2	AI Enquiry Assistant	First-line prospect question-answering and lead capture on public site	P1 (narrow)	CRM, Public Classes/Events data, LLM Gateway, Notifications
3	AI Booking Assistant	Shortlist eligible, available sessions for members; hand off to standard booking flow	P1 (narrow)	Classes, Bookings live availability, Memberships, LLM Gateway
4	AI Recommendation Assistant	Goal/level/history-based class suggestions filtered by eligibility and availability	P1 (narrow)	Classes, Memberships, Attendance, LLM Gateway
5	AI Feedback Assistant	Post-class free-text comment classification — sentiment, topic, theme summary for admins	P1 (narrow)	Classes, Attendance, LLM Gateway, Reports

Phase 2+ Agents (Future)

#	Module	Purpose	Priority	Key Dependencies
6	CRM Agent	Lead lifecycle management, scoring, draft follow-up messages for admin approval	P2	CRM, Notifications, Members, LLM Gateway
7	Operations Agent	Continuous capacity/attendance monitoring, alerts, schedule adjustment suggestions	P2	Classes, Bookings, Attendance, Memberships, LLM Gateway
8	Marketing Agent	Campaign drafting, audience segmentation, WhatsApp/email campaigns with human approval	P3	Members, CRM, Events, Notifications, LLM Gateway
9	Advertising Agent	Ad copy drafting, audience suggestions, performance analysis — human approval mandatory before any spend	P3	CRM, External Ad Platforms, LLM Gateway
10	Growth Optimization Agent	Revenue/occupancy analysis, pricing and schedule recommendations (advisory only)	P3	Reports, Classes, Memberships, LLM Gateway
11	Studio Insights Agent	Proactive business intelligence, trend summaries, anomaly detection, demand forecasts	P3	All modules read aggregates, LLM Gateway
12	Session Summary Agent	Attendance and approved trainer notes converted to readable member and staff summaries	F	Attendance, Classes, LLM Gateway

D Automation Modules (n8n Workflows)

#	Workflow	Purpose	Priority	Key Dependencies
1	Lead Automation	Lead registration confirmation WhatsApp; future follow-up sequence trigger	P1	CRM service, WhatsApp Cloud API, Notifications
2	Booking Automation	Booking confirmation WhatsApp; waitlist promotion notification	P1	Bookings service, WhatsApp Cloud API, Notifications

#	Workflow	Purpose	Priority	Key Dependencies
3	Class Reminder Automation	Class reminder WhatsApp 12 hours and 1 hour before session	P1	Classes service, Bookings service, WhatsApp Cloud API
4	Membership Renewal Automation	Renewal reminders at 7d, 3d, 1d before expiry and on expiry day	P1	Memberships service, WhatsApp Cloud API, Notifications
5	Payment Automation	Payment received confirmation WhatsApp immediately after confirmed payment	P1	Payments service, WhatsApp Cloud API, Notifications
6	Event Automation	Event registration confirmation; event reminders at 24h, 6h, 1h before event	P1	Events service, WhatsApp Cloud API, Notifications
7	Notification Sub-workflow	Shared send-message workflow: mock/live toggle, retry logic, delivery status storage	P1	WhatsApp Cloud API, Notifications module, PostgreSQL
8	Failure Alert Workflow	Surfaces persistent workflow failures to admin notification panel	P1	n8n error handling, Admin notification, Notifications module

AUTOMATION DESIGN PRINCIPLES

- All workflows consume signed webhook events from the Outbox Dispatcher
- Idempotency guaranteed by event ID — duplicate events produce no duplicate sends
- Phase 1 operates in mock/test mode; production credentials swap without code change
- Business logic never resides in n8n — only orchestration, recipient lookup, and template dispatch
- Every send attempt stored in Notification Log with recipient, template, status, timestamp

E ERP Modules (ERPNext on Frappe — Custom App: wellness_studio)

> **ARCHITECTURE RULE**
>
> ERPNext core is never modified. All product logic lives in the wellness_studio custom Frappe app. The ERP is the system of record for all business entities and executes all atomic transactional operations.

#	DocType /Module	Purpose	Priority	Key Dependencies
1	Member DocType	System-of-record member profile — name, phone (unique per tenant), email, status	P0	Linked to all wellness DocTypes
2	Membership Plan DocType	Studio-defined plans — name, price INR, validity, allowed class types, booking limits, active flag	P0	Membership assignment, Booking eligibility
3	Membership DocType	Member-plan assignment with state machine: Pending Approval → Active → Expired → Suspended → Cancelled	P0	Payments, Booking eligibility check
4	Class Type DocType	Configurable class categories — name, description, colour, active flag	P0	Class Session scheduling
5	Class Session DocType	Scheduled session — date, time, duration, trainer, class type, capacity, status, booking count	P0	Bookings atomic, Attendance

#	DocType /Module	Purpose	Priority	Key Dependencies
6	Booking DocType	Atomic booking record — member, session, status Booked/Waitlisted/Cancelled, queue position	P0	Class Session capacity, Member, Membership
7	Attendance Record DocType	Per-session attendance — member, session, method QR/Trainer/Admin, status, correction history	P0	Class Session, Booking, Member
8	Event DocType	Event record — name, category, date/time, capacity, audience type, price, registration window	P0	Payments if paid, Notifications
9	Event Registration DocType	Member/visitor registration — event, person, payment status, confirmation status	P0	Event, Member/Visitor, Payment Record
10	Lead DocType	Prospect record — name, phone, email, source, campaign, status pipeline, conversion link	P0	Member on conversion, CRM
11	Payment Record DocType	Internal payment — ID, provider Razorpay, provider ref, amount, currency, status, purpose, timestamps	P0	Membership, Event Registration, Gateway
12	Staff DocType	Employee/trainer — name, email, role assignment, studio scope, class assignments, active flag	P0	Roles, Class Session trainer reference
13	Tenant DocType	Tenant master — name, subdomain, tier Basic/ Premium/Ultra, feature flags, branding	P0	All DocTypes tenant scope
14	Studio DocType	Studio record — tenant, name, address, capacity defaults, cancellation window	P0	All session-level DocTypes
15	Branch DocType	Branch within studio — multi-location support from day one	P0	All DocTypes branch scope
16	Notification Log DocType	Delivery record — recipient, template, event type, status sent/failed/mock, timestamp, error	P1	Outbox, n8n, WhatsApp API
17	Audit Log DocType	Immutable audit — tenant, actor, actor type, action, resource, before/after values, timestamp	P0	All transactional DocTypes
18	AI Audit DocType	AI-specific immutable log — user, tool, arguments, decision, approval, result, timestamp	P1	AI Platform
19	Pending Action DocType	Proposed AI write-action — arguments fixed, requester, expiry, status pending/confirmed/expired	P1	AI Platform, Members service

SECTION 2

B2B Module Specifications

B2B-01 Dashboard

1 Module Overview

Home screen of the Studio Management Portal. Provides real-time operational summary: KPI tiles, renewal pipeline, today's schedule, quick actions, and AI Assistant access for Owners, Admins, and Front Desk.

2 Business Purpose

Eliminate navigation to multiple screens for daily questions: How many active members? Which memberships expire this week? What is today's attendance?

3 User Roles

Role	Access Level
Owner	Full view — KPIs, revenue tiles, renewal pipeline, AI Assistant
Admin	Full operational view within assigned studio, AI Assistant
Front Desk	Reduced — today's classes, check-in shortcut, lead capture
Trainer	Not used — uses Trainer Workspace screen

4 Features

- **KPI Tiles:** Active members, expiring this week, today's sessions, today's attendance, monthly revenue (Owner/Admin only)
- **Renewal Pipeline:** Members with expiry in 1-7 days with one-click renewal initiation
- **Today's Schedule:** Sessions with occupancy gauge and quick attendance link
- **Quick Actions:** Add Member, Record Payment, Capture Lead, Check In Member
- **Event Highlights:** Upcoming events in next 7 days
- **AI Assistant Panel:** Embedded inline (Owner/Admin when tier and feature flag permit)

5 Functions

- getDashboardSummary(tenantId, studioId, role, userId) — role-filtered KPI aggregates
- getRenewalPipeline(tenantId, studioId, daysAhead=7) — expiring memberships ordered by date
- getTodaysClasses(tenantId, studioId, date) — sessions with booking count and capacity
- getEventHighlights(tenantId, studioId, daysAhead=7) — upcoming events

6 User Actions

Action	Role	Result
View KPI tiles	Owner, Admin	Aggregated figures from reporting service

Action	Role	Result
Click renewal pipeline member	Owner, Admin	Navigate to membership detail for renewal
Click session in schedule	Owner, Admin, Front Desk	Navigate to session detail with attendance
Quick Action Add Member	Owner, Admin, Front Desk	Open Add Member form
Quick Action Record Payment	Owner, Admin, Front Desk	Open Record Payment modal
Quick Action Capture Lead	Owner, Admin, Front Desk	Open Lead Capture form
Quick Action Check In	Front Desk	Open check-in scanner/lookup

7 Business Rules

- Dashboard data is always scoped to user's tenant and studio — never cross-tenant
- KPI figures must match Reports module for same period and filters
- Memberships expiring within 7 days trigger renewal call-to-action
- Revenue tiles visible to Owner and Admin only
- AI Assistant panel renders only when tenant tier and feature flag both permit

8 Permissions Matrix

Capability	Owner	Admin	Front Desk	Trainer
View all KPI tiles	YES	YES	Partial no revenue	NO
View revenue tiles	YES	YES	NO	NO
View renewal pipeline	YES	YES	NO	NO
View today schedule	YES	YES	YES	NO
All quick actions	YES	YES	YES	NO
AI Assistant panel	YES	YES	Read only when enabled	NO

9 Validation Rules

- API responses must include tenant/studio scope match; mismatch returns 403
- KPI figures computed from same reporting service as Reports module
- Renewal pipeline cannot include members from other tenants

10 Workflow

```
User logs in
  JWT verified; tenant resolved from subdomain
  Role-aware dashboard assembled
  Parallel data requests (KPI, renewal pipeline, today's classes, event highlights)
  Each tile renders loading skeleton then populates
  On tile error: retry option shown; other tiles unaffected
```

11 Acceptance Criteria

- KPI totals reconcile with Reports module for same period
- Renewal pipeline shows all and only members expiring in 7 days or less
- Revenue tiles invisible to Front Desk (server-enforced)
- Every tile shows loading state then populated; error shows retry
- AI panel only visible when tier and feature flag active
- Dashboard loads within 3 seconds (NFR-PRF-02)

12 ERP Touchpoints

- Reporting service calls ERP aggregates for member counts, revenue, occupancy
- Renewal pipeline reads Membership DocType expiry dates
- Today's schedule reads Class Session DocType with booking counts

13 AI Touchpoints

- AI Assistant uses same reporting service as Dashboard tiles
- Identical data source guarantees AI answers match on-screen figures

14 Automation Touchpoints

- No n8n workflows triggered directly from Dashboard
- Renewal pipeline complements automated renewal reminder workflows

B2B-02 CRM Module

1 Module Overview

Manages the full prospect lifecycle: lead capture from any source, status pipeline, follow-up notes, duplicate detection, and conversion to member. Integrates with AI Enquiry Assistant for automated lead capture.

2 Business Purpose

Replace scattered WhatsApp messages and spreadsheet tracking with a governed lead pipeline. Measure acquisition ROI by source. Ensure every prospect gets timely follow-up. Preserve source attribution through to conversion.

3 User Roles

Role	Access
Owner	Full CRM — reports, analytics, conversion, source attribution
Admin	Create, read, update leads; mark conversion; view all records
Front Desk	Capture leads and enquiries; update status
Trainer	No CRM access
Visitor	Submit enquiry via public form or AI Enquiry Assistant

4 Features

- **Lead List:** Paginated, filterable by status, source, date range, assigned staff
- **Lead Detail:** Full profile, contact history, status, source/campaign, notes timeline
- **Lead Capture Form:** Name, phone, email, interest, source, campaign tag, consent
- **Status Pipeline:** New → Contacted → Trial Scheduled → Trial Done → Converted → Lost
- **Source Attribution:** Walk-in, WhatsApp, Facebook Ad, Instagram, Landing Page, Referral, Other (configurable)
- **Conversion Flow:** Convert lead to member; preserve source and campaign on member record
- **Duplicate Detection:** Flag leads with same phone or email within tenant

5 Functions

- createLead(tenantId, studioId, leadData) — ERP Lead DocType creation
- updateLeadStatus(leadId, newStatus, notes, actorId) — audited status transition
- addLeadNote(leadId, noteText, actorId) — append note to history
- convertLeadToMember(leadId, memberData, actorId) — atomic: create member, preserve source
- getLeads(tenantId, studioId, filters, pagination) — filtered list
- getLeadDetail(leadId, tenantId) — full profile with history

- detectDuplicates(phone, email, tenantId) — checks leads and members

6 User Actions

Action	Who	Result
Create new lead	Admin, Front Desk	Lead record in ERP; WhatsApp confirmation to lead
Update lead status	Admin, Front Desk	Status transitioned with timestamp
Add follow-up note	Admin, Front Desk	Note appended to timeline
Convert to member	Admin	Member created; lead status Converted; source preserved
View lead source report	Owner, Admin	Leads by source with conversion metrics
Mark lead as lost	Admin, Front Desk	Lead archived with reason
Merge duplicates	Owner, Admin	Duplicate leads merged

7 Business Rules

- BR-CRM-01: Advertisements and landing pages create Leads, never Members
- BR-CRM-02: Lead becomes Member only after conversion event; source and campaign preserved
- BR-CRM-03: Duplicate leads (same phone or email) flagged; admin must confirm merge before save
- BR-CRM-04: Lead follow-up WhatsApp messages subject to consent and approved template rules
- Lead source is set at creation and immutable
- Campaign tag editable by Admin only

8 Permissions Matrix

Capability	Owner	Admin	Front Desk	Trainer	Visitor
Create lead	YES	YES	YES	NO	Via form only
Update lead	YES	YES	YES	NO	NO
Convert to member	YES	YES	NO	NO	NO
Archive/delete lead	YES	YES	NO	NO	NO
View source reports	YES	YES	NO	NO	NO
Merge duplicates	YES	YES	NO	NO	NO

9 Validation Rules

- Phone: required; 10+ digits; numeric; unique within tenant (duplicate flag)
- Email: optional; valid format if provided
- Source: required; from configured source list in Settings
- Name: required; 2-100 characters

- Status transitions follow pipeline order; skipping requires Admin confirmation
- Consent checkbox required for WhatsApp follow-up

10 Workflow

```
Lead created (walk-in or landing page or AI Enquiry Assistant)
  Duplicate check on phone and email
  If duplicate: flagging modal (merge/proceed/cancel)
  Lead record created in ERP
  Outbox event: lead.created
  n8n lead-confirmation.json: WhatsApp confirmation to lead

Follow-up lifecycle:
  Admin contacts lead → Status: Contacted
  Trial arranged → Status: Trial Scheduled
  Trial done → Status: Trial Done
  Membership purchased → Conversion triggered
    Member record created in ERP with source preserved
    Lead status: Converted
  No conversion → Status: Lost (reason required)
```

11 Acceptance Criteria

- Lead creation creates ERP record; WhatsApp confirmation fires (mock mode Phase 1)
- Duplicate phone/email shows flagging modal before save
- Converted lead shows linked member record
- Lead source preserved on member record after conversion
- Lead list filterable by status, source, date range, assigned staff

12 ERP Touchpoints

- Lead DocType in custom Frappe app with source, campaign, status
- Conversion creates Member DocType linked to Lead ID
- Source and campaign stored on both DocTypes

13 AI Touchpoints

- AI Enquiry Assistant calls createLead via governed CRM service
- CRM Agent (Phase 2): reads leads, scores, drafts follow-up messages for admin approval

14 Automation Touchpoints

- n8n lead-confirmation.json fires on lead.created outbox event
- WhatsApp template: lead-registration-confirmation sent immediately

B2B-03 Members Module

1 Module Overview

Central identity and lifecycle hub for all studio members. Manages the complete profile, linked memberships, booking history, attendance history, and payment history.

2 Business Purpose

One trusted, searchable, duplicate-free record per member. Staff can onboard a member in under 1 minute and access their full history without switching systems.

3 User Roles

Role	Access
Owner	Full member management including archive
Admin	Create, read, update; approve memberships; audited corrections
Front Desk	Create, read, update permitted fields; cannot archive
Trainer	Read name, photo, eligibility for assigned-session participants only
Member	Own record only via B2C portal

4 Features

- **Member List:** Searchable by name, phone, email; filterable by status, plan, expiry
- **Member Detail:** Full profile plus linked memberships, bookings, attendance, payments tabs
- **Create Member Form:** Name (required), phone (required), email, date of birth, emergency contact, notes
- **Edit Member:** Permitted field updates with immutable audit trail
- **Duplicate Detection:** Real-time check on phone and email before save
- **Member Status Tags:** Active, Inactive, Expired, Suspended, No Membership
- **Quick Actions:** Book for member, Renew membership, Record payment, Send WhatsApp
- **Membership History Timeline:** All memberships in chronological order

5 Functions

- createMember(tenantId, studioId, memberData) — ERP Member DocType creation
- getMember(memberId, tenantId) — full profile with linked data
- getMembers(tenantId, studioId, filters, pagination) — list with search and filter
- updateMember(memberId, changes, actorId) — audited field update
- checkDuplicate(phone, email, tenantId) — pre-creation duplicate check
- getMemberHistory(memberId) — aggregated timeline
- archiveMember(memberId, reason, actorId) — soft-delete

6 User Actions

Action	Who	Result
Search members	All staff	Filtered member list
Create new member	Owner, Admin, Front Desk	Member record in ERP
View member detail	All staff (scoped)	Full or role-limited profile
Edit member fields	Owner, Admin, Front Desk	Immutable audit record created
Assign/renew membership	Owner, Admin	Membership created in ERP
Book on behalf	Owner, Admin, Front Desk	Booking via Bookings module
Record payment	Owner, Admin, Front Desk	Payment via Payments module
Archive member	Owner, Admin	Soft-archive; history preserved

7 Business Rules

- Member record never physically deleted — archived (status: Cancelled/Inactive)
- Duplicate check on phone is mandatory; on email is advisory
- Member must exist before membership can be assigned
- Members from other tenants never visible (server-side tenant isolation)
- Trainer sees only name, photo, eligibility for assigned-session participants — no financial data
- Phone number is member's primary login identifier — changes require identity re-verification

8 Permissions Matrix

Capability	Owner	Admin	Front Desk	Trainer	Member B2C
Create member	YES	YES	YES	NO	NO
Read full profile	YES	YES	YES	Assigned limited	Own only
Edit permitted fields	YES	YES	Limited	NO	Own permitted
Archive member	YES	YES	NO	NO	NO
View payment history	YES	YES	NO	NO	Own only
View attendance history	YES	YES	YES	Assigned sessions	Own only
Assign membership	YES	YES	NO	NO	Via purchase

9 Validation Rules

- Phone: required; 10+ digits; numeric; unique within tenant
- Name: required; 2-100 characters; no leading/trailing spaces
- Email: optional; valid format; advisory duplicate check

- Date of birth: optional; past date; 1900 to today
- Emergency contact: optional; valid phone format if phone provided
- Notes: optional; max 1000 characters

10 Workflow

```
Staff opens Add Member
  Studio and branch pre-filled from staff scope
  Staff enters name and phone
  Real-time duplicate check
  If duplicate: modal shows match (link/proceed)
  Staff completes form and submits
  API validates all fields
  ERP Member DocType created
  Optional: assign membership immediately
  Optional: welcome WhatsApp via outbox event
  Member appears in list with status from membership state
```

11 Acceptance Criteria

- Member creation completes in under 1 minute (operational KPI)
- Duplicate phone within tenant triggers modal before save
- All field edits produce audit record with actor, timestamp, before and after values
- Archived members retain full history; OTP login rejected for archived records
- Trainer view: name, photo, eligibility only — server-enforced
- Active member count on Dashboard matches module total

12 ERP Touchpoints

- Member DocType in custom Frappe app: system of record for member identity
- ERP enforces uniqueness on phone within tenant
- Membership, Attendance, Booking DocTypes reference Member by ERP document ID

13 AI Touchpoints

- AI Admin Assistant queries member counts, expiry lists via governed Members service
- AI Create Member action calls same createMember function with mandatory confirmation
- AI never accesses ERP Member DocType directly

14 Automation Touchpoints

- Member creation: optional member.created outbox event to n8n welcome WhatsApp
- Membership expiry scheduler drives renewal reminder automation
- No automated archival — all archiving is human-initiated

B2B-04 Memberships Module

1 Module Overview

Two-layer: (1) Studio-defined Member Plans (configurable, data-driven), and (2) Individual Membership Assignments (member-plan lifecycle with state machine). Gates class booking eligibility.

2 Business Purpose

Replace spreadsheet plan tracking with governed membership management. Enforce eligibility, manage approval, track expiry, drive renewal reminders, record every state change with full audit.

3 User Roles

Role	Access
Owner	Define/edit plans; approve/reject; full lifecycle management
Admin	Assign plans; approve (if granted); renew; view all
Front Desk	Create membership (Pending state only); cannot approve
Member	Browse plans; purchase; renew; view own (via B2C)

4 Features

PLAN MANAGEMENT

- Create, edit, deactivate studio-defined plans
- Plan fields: name, price INR, validity period, included class types, booking limits, active flag
- Deactivating plan does not retroactively affect active memberships

MEMBERSHIP ASSIGNMENT

- Assign plan to member with start date; end date computed from validity
- States: Pending Approval, Active, Expired, Suspended, Cancelled
- Payment and approval status are always independent fields

EXPIRY AND RENEWAL

- Expiry computed from start date plus plan validity
- Renewal reminders at 7d, 3d, 1d before expiry, and on expiry day
- Expiry does not auto-cancel; member status reflects expired membership

APPROVAL WORKFLOW

- Configurable: auto-approve on payment OR require manual admin approval
- Pending memberships block booking eligibility

5 Functions

- createPlan(tenantId, planData) — creates Plan in ERP (Owner only)
- updatePlan(planId, changes, actorId) — audited edit; no retroactive effect
- deactivatePlan(planId, actorId) — plan hidden from new purchases
- assignMembership(memberId, planId, startDate, actorId) — creates Membership
- approveMembership(membershipId, actorId) — Pending to Active
- suspendMembership(membershipId, reason, actorId) — Active to Suspended
- cancelMembership(membershipId, reason, actorId) — to Cancelled
- renewMembership(membershipId, actorId) — creates next-period membership
- getMembershipsForMember(memberId) — membership timeline
- getExpiringMemberships(tenantId, studioId, daysAhead) — renewal pipeline data

6 Business Rules

- BR-MSP-01: Studio plans are tenant-defined; platform tiers are separate and never mixed with member plans
- BR-MSP-02: States — Pending Approval, Active, Expired, Suspended, Cancelled. Payment and approval independent
- BR-MSP-03: End date must be after start date; price >= 0; member must exist
- BR-MSP-04: Phase 1 is manual renewal; recurring debit is a future opt-in capability
- BR-MSP-05: Member with Expired or Suspended membership cannot book restricted classes
- BR-MSP-06: Edits to active plan do not silently change existing memberships

7 Permissions Matrix

Capability	Owner	Admin	Front Desk	Member B2C
Create/edit plan	YES	NO	NO	NO
Deactivate plan	YES	NO	NO	NO
Assign membership	YES	YES	Pending only	Via purchase
Approve membership	YES	YES if granted	NO	NO
Renew membership	YES	YES	NO	Via purchase
Suspend/cancel	YES	YES	NO	NO
View all memberships	YES	YES	Own studio	Own only

8 Validation Rules

- Plan name: required; unique within tenant; max 100 characters
- Price: required; >= 0; INR default
- Validity: required; positive integer in days

- Start date: required; cannot be past unless Owner override
- Member: must exist; must belong to same tenant
- Approval cannot execute on unpaid membership unless plan is free

9 Workflow

```
Owner creates Plan (plan available for assignment and purchase)
Admin or Front Desk assigns Plan to Member with start date
  System computes end date from start date plus plan validity
  Membership created: state = Pending Approval

Member pays (online or manual)
  Payment confirmed server-side
  Auto-approval (if configured): Pending to Active immediately
  OR: Admin approves manually → Active
  Outbox event: membership.activated
  n8n: WhatsApp membership-activated sent

Renewal reminder schedule:
  7 days before expiry → WhatsApp renewal-reminder-7d
  3 days before expiry → WhatsApp renewal-reminder-3d
  1 day before expiry → WhatsApp renewal-reminder-1d
  Expiry day → WhatsApp membership-expired and state → Expired

Admin or Member renews
  New Membership record linked to previous
  Same flow: payment, approval, Active state
```

10 Acceptance Criteria

- Plan creation and deactivation do not affect existing active memberships
- Membership state transitions always produce audit record
- Member with Expired membership blocked from booking restricted classes (server-enforced)
- Renewal reminders fire at exactly 7d, 3d, 1d, and expiry day
- Payment confirmation and approval stored as independent state fields
- Expiring memberships <=7 days appear in Dashboard renewal pipeline

11 ERP Touchpoints

- Membership Plan DocType and Membership DocType in custom Frappe app
- ERP expiry scheduler fires callbacks to API for renewal pipeline queries
- Booking eligibility check: ERP validates membership state at booking commit time

12 AI Touchpoints

- AI Admin Assistant queries expiring memberships via governed Memberships service
- Recommendation Assistant uses plan entitlement data to filter class suggestions

13 Automation Touchpoints

- renewal-reminders.json n8n: daily schedule queries "what is due" from Memberships service
- Outbox events: membership.activated, membership.expiring, membership.expired
- WhatsApp templates: membership-activated, renewal-reminder-7d/3d/1d, membership-expired

B2B-05 Classes Module

1 Module Overview

Configurable class type definitions and scheduled class sessions. Manages trainer assignment, capacity, recurring schedules, and live availability.

2 Business Purpose

Replace spreadsheet schedules with a governed system that enforces capacity, prevents conflicts, tracks trainer assignments, and provides live availability to members.

3 User Roles

Role	Access
Owner	Full class type and session management
Admin	Full session management within assigned studio
Front Desk	View schedule; book on behalf of members
Trainer	View own assigned sessions only
Member	View available schedule (filtered by plan entitlement)

4 Features

- **Class Types:** Fully configurable — Yoga, Zumba, Dance, Meditation, Pilates (all data-driven, none hard-coded)
- **Session Scheduling:** Date, time, duration, trainer, room, class type, capacity (default 30)
- **Recurring Schedules:** Weekly recurring series with per-instance override
- **Trainer Assignment:** One trainer per session; can be reassigned
- **Capacity Tracking:** Real-time booking count vs. capacity; waitlist count
- **Session Status:** Scheduled, Active, Completed, Cancelled
- **Session Detail:** Full info, enrolled participants, attendance status

5 Functions

- createClassType(tenantId, classTypeData) — new configurable class category
- createSession(tenantId, studioId, sessionData) — one-time or recurring session
- updateSession(sessionId, changes, actorId) — audited edit
- cancelSession(sessionId, reason, actorId) — cancel; notify enrolled members
- getSessionSchedule(tenantId, studioId, dateRange, filters) — schedule view
- getSessionDetail(sessionId, tenantId) — full session with participants
- getTrainerSessions(trainerId, tenantId, dateRange) — trainer-scoped view

6 Business Rules

- Default capacity is 30; configurable per class type and per session
- Session capacity cannot be reduced below current booking count (server-enforced)
- Cancelling a session releases all bookings; waitlisted members are notified not auto-promoted
- Trainer sees only sessions assigned to them
- Class type cannot be deleted if active sessions reference it — deactivate only
- No hard-coded class type names (BR-GEN-05)

7 Permissions Matrix

Capability	Owner	Admin	Front Desk	Trainer	Member
Create class type	YES	YES*	NO	NO	NO
Create session	YES	YES	NO	NO	NO
Edit session	YES	YES	NO	NO	NO
Cancel session	YES	YES	NO	NO	NO
View full schedule	YES	YES	YES	NO	Filtered view
View assigned sessions	YES	YES	YES	Assigned only	NO
View participant list	YES	YES	YES	Assigned limited	NO

8 Validation Rules

- Session date: required; cannot be past unless Owner override
- Session time: required; valid 24h format
- Duration: required; positive integer in minutes
- Trainer: required; active staff in same studio
- Capacity: required; positive integer; defaults to studio default
- Class type: required; active class type in same tenant
- Recurring end date: if set, must be after first session date

9 Workflow

```
Admin creates class type if new vertical
Admin creates session (or recurring series)
  Sets date, time, duration, trainer, class type, capacity
  System generates all recurring instances if repeating
Sessions visible in member-facing schedule
Session day:
  QR code generated
  Members attend and check in
  Trainer marks attendance post-session
  Session status → Completed
```

10 Acceptance Criteria

- Class types entirely data-driven; no class name hard-coded in codebase
- Session creation with recurrence generates all instances correctly
- Cancelling session notifies all booked members via WhatsApp
- Trainer-scoped view contains only assigned sessions (server-enforced)
- Capacity reduction below booking count rejected at server level
- Schedule view loads within 3 seconds for a month of sessions

11 ERP Touchpoints

- Class Type DocType: name, description, colour, active flag
- Class Session DocType: all session fields, trainer, capacity, booking count, status
- Booking DocType references Class Session for eligibility and capacity

12 AI Touchpoints

- AI Booking Assistant queries live session availability via Classes service
- AI Recommendation Assistant uses class type and schedule data
- Operations Agent (Phase 2) reads occupancy to detect low-attendance classes

13 Automation Touchpoints

- session.cancelled outbox event → n8n notification to all enrolled members
- class-reminders.json: n8n sends reminder 12h and 1h before session

B2B-06 Bookings Module

1 Module Overview

Manages the complete lifecycle of class seat reservations. Handles eligibility validation, atomic capacity enforcement, waitlist management, cancellations, and staff booking-on-behalf.

2 Business Purpose

Prevent overbooking, enforce membership eligibility, automatically manage the waitlist, and ensure every booking decision is authoritative and atomic.

3 User Roles

Role	Access
Owner	Full booking management; capacity override
Admin	Book, cancel, view all bookings within studio
Front Desk	Book and cancel on behalf of members
Trainer	View own session bookings (if permission explicitly granted)
Member	Self-book; cancel within cutoff; view own bookings

4 Features

- **Live Availability:** Real-time seats available and waitlist count per session
- **Eligibility Check:** Active membership, plan entitlement, booking limits
- **Atomic Booking:** Server-side lock prevents last-seat overbooking (NFR-PRF-03)
- **Waitlist:** Automatic promotion when seat opens
- **Cancellation:** Self-service within cutoff (default 4 hours); admin anytime
- **Book on Behalf:** Admin and Front Desk can book for any member
- **Booking Status:** Booked, Waitlisted, Cancelled, Attended, No-Show

5 Functions

- bookSession(memberId, sessionId, actorId, onBehalf) — atomic ERP booking with all checks
- cancelBooking(bookingId, actorId, reason) — cancels booking; triggers waitlist promotion
- getAvailability(sessionId) — current seats available and waitlist count
- getBookingsForSession(sessionId, tenantId) — all bookings for session
- getBookingsForMember(memberId, tenantId, filters) — member booking history
- promoteWaitlist(sessionId) — promotes next eligible waitlisted member (atomic ERP call)
- bookOnBehalf(staffId, memberId, sessionId) — staff-initiated booking

6 Business Rules

- BR-BKG-01: Backend and ERP perform final validation at commit — UI cannot bypass
- BR-BKG-02: Default capacity 30; configurable per class type and per session
- BR-BKG-03: Default cancellation cutoff 4 hours; configurable
- BR-BKG-04: Waitlist enabled; next eligible waitlisted member promoted automatically in queue order
- BR-BKG-05: Members self-book; Admin and FD book on behalf; Trainers assist only when explicitly granted
- BR-BKG-06: No role may bypass membership eligibility, payment, or capacity rules
- Member cannot have two bookings for the same session
- Booking confirmation WhatsApp sent only after booking committed

7 Permissions Matrix

Capability	Owner	Admin	Front Desk	Trainer	Member
Self-book	YES	YES	YES	YES	YES
Book on behalf	YES	YES	YES	Granted only	NO
Cancel any booking	YES	YES	YES	NO	Own within window
View session bookings	YES	YES	YES	Assigned granted	NO
Override cancellation window	YES	YES	NO	NO	NO
Force-book past capacity	YES	NO	NO	NO	NO

8 Validation Rules

- Member must exist and belong to same tenant
- Session must be Scheduled (not Cancelled or Completed)
- Seats available OR waitlist enabled
- Member must have Active membership entitling them to the class type
- Member must not already have a booking for this session
- Booking must be created before session start time
- Member self-cancellation must be before configured cutoff

9 Workflow

```
Member or Staff initiates booking
  API validates: JWT + tenant scope + session exists + session not cancelled
  ERP atomic call with row lock:
    Check eligibility: membership active and class type entitled
    Check booking limit: member has not exceeded plan limit
    Check duplicate: member not already booked for this session
    Check capacity: seats > 0?
      YES → Create Booking (status: Booked)
```

```
      NO → Waitlist enabled?
        YES → Create Booking (status: Waitlisted)
        NO → Return Session Full
  Audit record written (same transaction)
  Outbox event: booking.confirmed or booking.waitlisted
  Response returned to UI (authoritative result)
  n8n booking-confirmation.json: WhatsApp confirmation sent

Cancellation:
  Member or Staff cancels
  Member self-cancel: check cutoff window (4h default)
  Booking status → Cancelled
  Outbox event: booking.cancelled
  n8n triggers promoteWaitlist:
    Next eligible waitlisted booking → Booked
    WhatsApp: seat-promotion-confirmation to promoted member
```

10 Acceptance Criteria

- Last-seat contention: exactly one request succeeds (NFR-PRF-03)
- Ineligible class shows reason: "Your plan does not include this class type"
- Confirmation reflects authoritative ERP result not cached state
- Cancellation cutoff shown and enforced; configurable; default 4 hours
- Waitlist promotion automatic; promoted member notified via WhatsApp
- Booking response time: 95th percentile under 1 second (NFR-PRF-01)

11 ERP Touchpoints

- Booking DocType: atomic booking with row-locking in custom Frappe app API method
- ERP enforces eligibility, capacity, duplicate in one transaction
- Session DocType holds booking count; updated atomically
- Waitlist records in ERP with queue position; promotion is atomic ERP method call

12 AI Touchpoints

- AI Booking Assistant uses getAvailability and eligibility to present shortlist
- AI never calls bookSession directly — member confirms through standard flow

13 Automation Touchpoints

- booking-confirmation.json n8n fires on booking.confirmed outbox event
- class-reminders.json n8n fires 12h and 1h before session for all booked members
- booking.promoted outbox event triggers WhatsApp notification to promoted member

B2B-07 Attendance Module

1 Module Overview

Records, corrects, and reports member attendance for class sessions and events. Three capture methods: QR check-in, trainer marking, audited admin correction.

2 Business Purpose

Replace paper registers with digital, tamper-evident attendance records that are accurate, analysable, and auditable for every session.

3 User Roles

Role	Access
Owner	View all; correct any attendance with audit
Admin	Mark and correct attendance; view all within studio
Front Desk	QR check-in; view session attendance
Trainer	Mark attendance for assigned sessions only
Member	QR self-check-in; view own history

4 Features

- **QR Check-In:** Member scans session QR from B2C app or printed sheet
- **Trainer Marking:** Marks each participant Present/Absent post-session
- **Admin Correction:** Audited correction with mandatory reason; original preserved
- **Session Roster:** Booked vs. Attended comparison per session
- **Member Attendance History:** All past sessions with status

5 Functions

- checkIn(memberId, sessionId, method, actorId) — idempotent; validates booking and no-duplicate
- markAttendance(sessionId, attendanceList, trainerId) — trainer bulk marking (assigned sessions only)
- correctAttendance(attendanceId, newValue, reason, actorId) — immutable correction record
- getSessionAttendance(sessionId, tenantId) — full roster with status
- getMemberAttendanceHistory(memberId, tenantId, filters) — member's past attendance
- generateSessionQR(sessionId, tenantId) — time-limited session-specific QR token

6 Business Rules

- BR-ATT-01: Attendance recorded per class session or event session, not per day

- BR-ATT-02: Supported methods — QR check-in, trainer marking, audited admin correction
- BR-ATT-03: Every correction records actor, timestamp, previous value, new value, reason; entry is immutable
- BR-ATT-04: Member cannot have two attendance records for the same session
- BR-ATT-05: Trainers may mark attendance only for sessions assigned to them
- QR token is session-specific and time-limited
- Check-in valid only for members with a confirmed Booking for the session
- Correction reason is mandatory

7 Permissions Matrix

Capability	Owner	Admin	Front Desk	Trainer	Member
QR check-in	YES	YES	YES	NO	YES self
Manual check-in	YES	YES	YES	NO	NO
Bulk mark attendance	YES	YES	NO	Assigned only	NO
Correct attendance	YES	YES	NO	NO	NO
View session roster	YES	YES	YES	Assigned	NO
View member history	YES	YES	YES	Assigned members	Own only

8 Workflow

```
Session starts → QR code generated and displayed

Path A — QR Check-in:
  Member opens B2C app → Scan QR
  API validates: token validity + member has active booking + no duplicate
  Attendance record created (method: QR, status: Present)
  Confirmation shown in app

Path B — Trainer Marking:
  Post-session: Trainer opens assigned session
  Views participant list with Booked status
  Marks each member Present or Absent
  Bulk save → Attendance records created
Path C — Admin Correction:
  Admin finds attendance record
  Selects Correct → Enter new value and mandatory reason
  Correction record created (original preserved and correction linked)
  Correction in Audit Log with full attribution
```

9 Acceptance Criteria

- Duplicate check-in returns "Already checked in" — no duplicate record created
- Correction creates immutable audit record; original preserved
- Trainer blocked from marking attendance for unassigned sessions (server-enforced)
- QR token expiry validated server-side

- Session roster shows Booked vs. Attended counts
- Attendance capture completeness: 98%+ of sessions (operational KPI)

10 ERP Touchpoints

- Attendance Record DocType: member, session, method, status, timestamp
- Correction stored as child record with immutable history
- ERP enforces no-duplicate constraint per member per session

11 AI Touchpoints

- AI Admin Assistant queries "low attendance classes" via Attendance/Reports service
- Feedback Assistant correlates attendance records with post-class comments

12 Automation Touchpoints

- No-show processing (Phase 2): n8n detects booked-not-attended after session window closes
- Phase 2: inactive-member nudge uses attendance history

B2B-08 Events Module

1 Module Overview

Manages workshops, studio events, public events, and special sessions. Events are distinct from regular classes with their own capacity, registration window, audience type, pricing, and notification sequence.

2 Business Purpose

Enable revenue-generating workshops and engagement events that attract new members and re-engage existing ones.

3 User Roles

Role	Access
Owner	Full event management
Admin	Create, edit, manage events within studio
Front Desk	Register members for events; view registrations
Trainer	View assigned events
Member	Browse and register for eligible events
Visitor	Browse public events; register for public workshops

4 Features

- **Event Categories:** Workshop, Internal Studio Event, Public Event, Special Session (all configurable)
- **Event Config:** Name, description, date/time, duration, capacity, price or free, audience type, registration window
- **Audience Types:** Open to all, Members only, Prospects only (configurable)
- **Registration Management:** Registrations, seat count, payment status, waitlist
- **Payment Integration:** Paid events require verified payment before confirmation
- **Notification Sequence:** Registration confirmation; reminders 24h, 6h, 1h before

5 Functions

- createEvent(tenantId, studioId, eventData) — new Event DocType in ERP
- updateEvent(eventId, changes, actorId) — audited edit
- cancelEvent(eventId, reason, actorId) — cancel; notify all registrants
- registerForEvent(registrantId, eventId, actorId) — registration with eligibility and capacity check
- getEvents(tenantId, studioId, filters) — event list (public or staff view)
- getEventRegistrations(eventId, tenantId) — registrant list with payment status

6 Business Rules

- BR-EVT-01: Default categories configurable; Workshop, Internal Studio Event, Public Event, Special Session
- BR-EVT-02: Events may be free or paid; open to all, members only, or prospects only
- BR-EVT-03: Each event stores capacity, date/time, registration window, price, audience type
- BR-EVT-04: Paid registration confirmed only after verified server-side payment; seat limits always enforced
- Members-only events hidden from Visitors in public view
- Event capacity enforcement is atomic (same pattern as class booking)

7 Permissions Matrix

Capability	Owner	Admin	Front Desk	Trainer	Member	Visitor
Create event	YES	YES	NO	NO	NO	NO
Edit event	YES	YES	NO	NO	NO	NO
Cancel event	YES	YES	NO	NO	NO	NO
Register others	YES	YES	YES	NO	NO	NO
Self-register	YES	YES	YES	YES	YES	Public events
View registrations	YES	YES	YES	Assigned	NO	NO

8 Workflow

```
Admin creates Event (audience type determines visibility)
Member or Visitor discovers event and clicks Register
  Eligibility check (audience type vs. user type)
  Seat availability check (atomic ERP)

  Free event:
    Registration confirmed immediately
    n8n: WhatsApp event-registration-confirmation

  Paid event:
    Razorpay checkout initiated
    Server-side webhook validates payment
    Registration confirmed after verified payment
    n8n: WhatsApp event-registration-confirmation and payment-received

Event reminders: 24h, 6h, 1h before event (n8n scheduled)
Event day: Attendance recorded via Attendance module
```

9 Acceptance Criteria

- Members-only events invisible to Visitors (server-enforced)
- Paid event registration confirmed only after server-verified payment

- Event capacity enforced atomically
- Event cancellation notifies all registrants via WhatsApp
- Registration window enforced; no registrations after cutoff

10 ERP Touchpoints

- Event DocType: all event fields, capacity, audience type
- Event Registration DocType: registrant, payment status, confirmation status
- Payment Record DocType linked to Event Registration for paid events

11 AI Touchpoints

- AI Enquiry Assistant answers questions about upcoming events and assists with registration
- Marketing Agent (Phase 3) uses event data to draft promotional campaigns

12 Automation Touchpoints

- event-registration.json n8n: fires on event.registration.confirmed outbox event
- Event reminder n8n schedule: 24h, 6h, 1h before event start
- event.cancelled outbox event triggers cancellation notification workflow

B2B-09 Payments Module

1 Module Overview

Handles online payment via Razorpay (test mode Phase 1), manual offline payment recording, payment status tracking, and reconciliation.

2 Business Purpose

Ensure every payment is recorded, server-side verified, accurately tracked, and reconcilable. Eliminate cash reconciliation errors and payment status disputes.

3 User Roles

Role	Access
Owner	Full payment visibility; refund authority
Admin	Record and view payments; refund if granted
Front Desk	Record manual/cash payments; view limited info
Member	Own payment history and checkout
Trainer	No payment access

4 Features

- **Online Payment:** Razorpay checkout (test mode Phase 1); production swap without code change
- **Payment Record Fields:** Internal ID, provider, provider reference, amount, currency INR default, status, purpose, timestamps
- **Manual Payment Recording:** Cash, bank transfer, UPI — with actor attribution
- **Payment Status:** Created, Pending, Paid, Failed, Refunded
- **Exception List:** Failed and mismatched payments visible to Admin

5 Functions

- initiatePayment(tenantId, memberId, purpose, amount, metadata) — creates Razorpay order
- handlePaymentCallback(paymentId, razorpayData, signature) — server-side webhook verification
- recordManualPayment(tenantId, memberId, amount, method, reference, actorId) — offline payment
- getPayments(tenantId, filters, pagination) — full payment list
- getPaymentsForMember(memberId, tenantId) — member's own payment history

6 Business Rules

- BR-PAY-01: Payment gateway is Razorpay; test mode until production credentials supplied
- BR-PAY-02: Only verified server-side gateway confirmation marks payment as Paid

- BR-PAY-03: Payment confirmation and membership approval are independent states
- BR-PAY-04: Each payment records internal ID, provider, provider reference, amount, currency, status, purpose, timestamps
- BR-PAY-05: Gateway callbacks are idempotent; delayed callback leaves payment unresolved
- BR-PAY-06: Refunds and manual adjustments require permission and are audited
- Failed payment never activates membership or confirms event registration

7 Permissions Matrix

Capability	Owner	Admin	Front Desk	Member
Initiate online payment	YES	YES	YES	YES own
Record manual payment	YES	YES	YES	NO
View all payments	YES	YES	Limited	Own only
Process refund	YES	If granted	NO	NO
View reconciliation	YES	YES	NO	NO

8 Workflow

```
Online Payment:
  Member or Staff initiates payment
  API creates Razorpay order
  Payment record created (status: Pending)
  Checkout link returned to frontend
  Member completes payment on Razorpay
  Razorpay fires signed webhook to API
  API verifies signature using Razorpay secret
  Payment status updated: Paid or Failed
  If Paid:
    Outbox event: payment.confirmed
    Membership activated OR Event registration confirmed
    n8n payment-received.json: WhatsApp payment-received sent
  If Failed:
    Payment remains Failed
    Admin exception list updated

Manual Payment:
  Front Desk or Admin records cash/UPI payment
  Payment record created with actor attribution
  Membership activation triggered if applicable
  WhatsApp: payment-received sent
```

9 Acceptance Criteria

- Server-side Razorpay signature verification mandatory; no payment marked Paid without it
- Duplicate webhook callbacks produce no duplicate payment records
- Failed payment never activates membership or confirms event registration
- Manual payment records include actor attribution; cannot be edited after creation
- Payment confirmation WhatsApp fires only after payment status confirmed Paid

10 ERP Touchpoints

- Payment Record DocType in ERP custom app
- ERP accounting structures used in Phase 2 for GST/invoicing

11 AI Touchpoints

- AI cannot initiate or modify payments — fully deterministic business logic
- Reports module provides revenue figures for AI queries

12 Automation Touchpoints

- payment-received.json n8n fires on payment.confirmed outbox event
- Failed payment surfaced to admin exception list

B2B-10 Reports Module

1 Module Overview

Provides operational and financial insight through pre-built report views. Uses same governed reporting service as Dashboard and AI Assistant — guaranteeing identical figures.

2 Phase 1 Core Reports

Report Name	Description	Role Access
Active Members	Count and list by plan	Owner, Admin
Revenue by Plan	Monthly/period revenue by plan	Owner, Admin
Class Occupancy	Bookings vs. capacity per session	Owner, Admin
Attendance Trend	Attended vs. booked over time	Owner, Admin
Renewal Pipeline	Members expiring in 7/14/30 days	Owner, Admin
Lead Conversion	Leads by source and conversion status	Owner, Admin

3 Business Rules

- Report data always scoped to requesting user's tenant and studio
- Reports use same data source as Dashboard KPI tiles
- Access respects role scope; Front Desk cannot see revenue reports (server-enforced)

4 Acceptance Criteria

- Report totals reconcile with Dashboard tiles for same period
- Revenue report accessible only to Owner and Admin (server-enforced)
- Reports load within 3 seconds
- Same metric and period give identical figures in Dashboard, Reports, and AI Assistant

B2B-11 Staff Management Module

1 Module Overview

Manages creation, role assignment, permission grants, and deactivation of all B2B users. Platform's security and access control hub.

2 Business Purpose

Enforce least-privilege security. Prevent unauthorized access to financial data, member records, or configuration.

3 Features

- **Staff List:** All staff with role, studio scope, status, last login
- **Invite Staff:** Email-based invitation with role pre-selection
- **Role Assignment:** Owner, Admin, Front Desk, Trainer
- **Granular Permission Grants:** Specific capability overrides
- **Trainer Profiles:** Specialty, assigned class types, bio
- **Deactivation:** Revokes auth access; historical records retain attribution

4 Business Rules

- Role changes take effect on next authenticated request
- Deactivated users cannot authenticate; historical attribution records preserved
- Owner cannot deactivate themselves if last Owner for tenant
- Trainer access to financial data cannot be granted regardless of any permission configuration
- All role changes and permission grants are immutable audit records
- MFA enforcement for Owner and Admin before production release (Phase 2)

5 Permissions Matrix

Capability	Owner	Admin
Invite staff any role	YES	Front Desk and Trainer only
Assign Owner/Admin role	YES	NO
Assign Front Desk/Trainer role	YES	YES
Grant granular permissions	YES	NO
Deactivate staff	YES	Front Desk and Trainer only

6 Acceptance Criteria

- Role changes effective on next authenticated request
- Deactivated staff cannot authenticate
- Historical records retain attribution after deactivation
- Trainer cannot access financial data regardless of permission grant

B2B-12 Settings (Configuration Center)

1 Module Overview

Externalizes all business rules, branding, templates, and feature configuration. No developer deployments needed for business rule changes.

2 Phase 1 Configuration Sections

Section	What is Configured	Role
Studio Profile	Name, logo, colours, contact, address	Owner
Member Plans	Create/edit/deactivate studio plans	Owner
Class Types	Configure all class categories	Owner, Admin
Default Capacity	Studio-wide default session capacity	Owner
Cancellation Window	Default cutoff for member self-cancellation	Owner
Event Categories	Configure event types	Owner, Admin
Notification Templates	WhatsApp template text and timing	Owner, Admin
Reminder Timings	Configurable windows for all notification types	Owner, Admin
Feature Flags	Enable/disable AI and WhatsApp automation	Owner

3 Business Rules

- Configuration changes take effect without deployment
- All configuration changes are immutable audit records
- New wellness vertical requires configuration only — no hard-coded names (BR-GEN-05)
- Deactivating a plan does not retroactively affect existing active memberships
- Secrets are write-only in UI — never displayed after saving

4 Acceptance Criteria

- Changing studio logo updates member portal theme without redeployment
- Changing reminder timing updates subsequent sends without corrupting existing records
- All configuration changes appear in audit log
- New class type added immediately available for session scheduling

B2B-13 Notifications Module

1 Module Overview

Manages all WhatsApp communication: template management, delivery status, mock/live mode, admin failure visibility. Integrates with n8n and WhatsApp Cloud API.

2 Notification Templates and Timing

Event	Template	Default Timing
Lead created	`lead-registration-confirmation`	Immediately
Membership activated	`membership-activated`	Immediately
Payment received	`payment-received`	Immediately after verified payment
Booking confirmed	`booking-confirmation`	Immediately
Class reminder	`class-reminder`	12h and 1h before session
Event registration	`event-registration-confirmation`	Immediately
Event reminder	`event-reminder`	24h, 6h, 1h before
Workshop reminder	`workshop-reminder`	24h, 12h, 1h before
Renewal reminder	`renewal-reminder`	7d, 3d, 1d before expiry
Membership expired	`membership-expired`	On expiry day

All timings configurable in Settings.

3 Business Rules

- BR-NTF-01: n8n orchestrates; WhatsApp Cloud API delivers; status stored per message
- BR-NTF-02: Messages sent only after transaction commits; failed message never reverses valid transaction
- BR-NTF-03: Each event plus recipient produces at most one message (idempotency by event ID)
- BR-NTF-04: Until production account exists, test/mock mode used; outbound records stored
- BR-NTF-05: Only approved templates used; marketing messages require opt-in; opt-outs honoured

4 Acceptance Criteria

- Mock mode stores one would-send record per message — admin-inspectable
- Same event plus recipient never produces two WhatsApp sends
- Failed notification never reverses business transaction
- Delivery status updated when WhatsApp API callback received

5 Automation Touchpoints

- All notification workflows in n8n: booking-confirmation.json, class-reminders.json, renewal-reminders.json, payment-received.json, event-registration.json, lead-confirmation.json
- Shared send-message.json sub-workflow: mock/live toggle, retry, status update
- failure-alert.json: surfaces persistent send failures to admin

B2B-14 Tenant Management

1 Module Overview

Operates the platform as a multi-tenant SaaS. Provides structural isolation per studio tenant, subscription tier hierarchy, and Platform Administrator onboarding.

2 Tenant Hierarchy

Tenant → Studio(s) → Branch(es)

3 Features

- Data isolation: every record scoped by tenant; cross-tenant access denied and logged
- Subscription tiers: Basic, Premium, Ultra with feature entitlement flags
- Feature flags: AI, WhatsApp automation gated by tier
- Subdomain routing: tenant resolved from subdomain

4 Business Rules

- Every business record is tenant-scoped; cross-tenant access denied and logged (BR-GEN-02)
- Tier flags gate AI and automation features per tenant (BR-AI-11)
- Tenant, studio, and branch structure is data-modelled from day one (NFR-SCL-01)
- Platform Administrator has no routine access to tenant business data except through audited support procedures

5 Acceptance Criteria

- Cross-tenant data access denied at server level and logged
- Tier change updates feature entitlements immediately
- Onboarding creates: Tenant, Studio, Branch, Owner user account
- All business records include tenant_id; cannot be queried without matching tenant context

B2B-15 AI Assistant Module (B2B)

1 Module Overview

Admin-facing natural-language interface embedded in the Studio Management Portal. Enables staff to query operational data and perform Create Member action conversationally through the same governed capabilities as the standard UI.

2 Phase 1 MVP Scope

- **Read Query 1:** Which memberships expire this week?
- **Read Query 2:** Which classes have low attendance?
- **Write Action:** Create Member with field extraction, confirmation, and execution

3 User Roles

Role	Access
Owner	All queries and Create Member action
Admin	All queries and Create Member action
Front Desk	Read queries only (when Owner enables)
Trainer	No AI Assistant access

4 Business Rules

- BR-AI-01: AI acts only through governed capabilities; never reads or writes database directly
- BR-AI-02: AI inherits signed-in user's permissions and tenant scope
- BR-AI-03: Transactional actions require explicit user confirmation before execution
- BR-AI-04: Assistant never guesses missing mandatory data — asks for it
- BR-AI-05: Assistant reports actual result — never claims success not received
- BR-AI-06: Every AI action logged with initiator, capability, arguments, result, timestamp
- BR-AI-07: Conversation context is temporary; becomes business data only through governed capability
- BR-AI-11: AI features gated by subscription tier and feature flag per tenant

5 Workflow

```
Admin types message in chat panel
  Message sent to AI Orchestrator with: session, user identity, role, tenant
  Orchestrator builds system prompt with permitted tool definitions only
  LLM called via AI Gateway

  Path A — Read Query:
    LLM selects read tool (e.g., getExpiringMemberships)
    Tool called via domain service
    LLM summarises ONLY from returned data
```

```
    Concise answer displayed

Path B — Create Member:
  LLM recognises Create Member intent
  Extracts fields: name, plan, start date
  Missing mandatory fields → Prompt user one at a time
  All fields collected → Confirmation card displayed:
    "Create member: Rahul Kumar | Plan: Premium | Starting: 30 Sep 2026
     Confirm? [Yes] [Cancel]"
  User confirms:
    Stored pending action arguments executed via createMember capability
    Real result reported truthfully (success with ID or error message)
  AI Audit record written (immutable)
```

6 Acceptance Criteria

- At least 85% of curated test queries answered correctly without escalation
- At least 95% of confirmed Create Member actions succeed on first confirmation
- Zero false-success reports — ERP error reported truthfully
- AI answers identical to Dashboard figures for same period and user scope
- User without Create Member permission cannot execute action via Assistant (server-enforced)
- Relative dates (Monday) resolved and shown in confirmation summary
- Unsupported questions receive clear scope refusal — no hallucinated answer
- Every call produces immutable AI Audit record

SECTION 3

B2C Module Specifications (Member Portal)

B2C-01 Member Dashboard

1 Module Overview

The personalised home screen of the Member Portal. Gives members a clear, immediate view of their membership status, upcoming bookings, renewal urgency, and a nudge toward next action.

2 Business Purpose

Reduce member support queries by surfacing the three questions members ask most: "Is my membership still active?", "What is my next class?", and "How do I renew?"

3 Features

- **Membership Status Card:** Plan name, validity dates, status badge (Active / Expiring Soon / Expired), progress indicator
- **Upcoming Bookings Panel:** Next 3 upcoming class sessions with date, time, class type, trainer, cancellation shortcut
- **Renewal Call-to-Action:** Sticky banner when membership expires within 7 days or is already expired; links to Membership Purchase flow
- **Quick Actions:** Book a Class, Browse Events, View Attendance
- **Recent Attendance Strip:** Last 3 sessions attended with date and class type

4 User Roles

Role	Access
Authenticated Member	Own data only — no other member's data visible at any point
Unauthenticated Visitor	Redirected to login; cannot access any member portal screen

5 Functions

- getMemberDashboard(memberId, tenantId) — aggregates from Memberships, Bookings, Attendance services
- Returns: active membership, next N bookings, last N attended sessions, expiry urgency flag

6 Validation Rules

- Session must validate JWT + tenant context before returning any data
- Expired session returns 401 → redirect to login
- Data returned is always scoped to the authenticated member's ID — never accepts memberId from URL params without matching JWT claim

7 Business Rules

- Dashboard data is always scoped to the authenticated member's tenant and identity
- Renewal CTA visible whenever membership expires within 7 days or is Expired
- Membership card status badge reflects real-time ERP state — not a cached value

8 Acceptance Criteria

- Dashboard loads within 3 seconds (NFR-PRF-02)
- All data belongs to the authenticated member — no cross-member data leakage possible
- Renewal CTA appears when membership expires in ≤7 days or is already Expired
- Expired membership shows Expired badge; member cannot book classes from expired membership
- Quick actions navigate correctly to respective modules

B2C-02 Membership Module

1 Module Overview

Allows members to browse studio membership plans, purchase a new membership, renew an existing one, and view their full membership history.

2 Business Purpose

Remove friction from membership purchase and renewal. Let members self-serve completely — browse plans, pay online, and get instant confirmation — without needing to visit the studio or call staff.

3 Features

- **Plan Browsing:** All active plans with name, price, validity, included class types, and booking limits
- **Plan Comparison:** Side-by-side comparison of available plans
- **Purchase Flow:** Select plan → Set start date → Razorpay checkout → Confirmation
- **Renewal Flow:** Renew current plan or switch to different plan
- **Membership History:** Chronological list of all past and current memberships with dates, plan, status
- **Current Membership Detail:** Active plan, start date, end date, days remaining, included class types

4 User Roles

Role	Access
Authenticated Member	Own memberships only
Unauthenticated Visitor	Can view plans (public-facing); purchase requires login

5 Functions

- getActivePlans(tenantId) — all active studio plans visible to members
- getMemberMemberships(memberId, tenantId) — member's full membership history
- purchaseMembership(memberId, planId, startDate) — initiates Razorpay checkout; creates Pending membership
- renewMembership(memberId, existingMembershipId, planId) — creates renewal linked to previous

6 Business Rules

- Plans displayed are only those marked active by the studio (Admin-controlled in Settings)
- Purchase creates Membership record in Pending state; activation occurs after payment verified
- Start date can be today or a future date; past dates not permitted
- Member can hold only one Active membership at a time (additional plan requires approval or auto-stacking policy configured by Owner — Phase 2)
- Renewal creates new Membership record; does not extend existing record

7 Validation Rules

- Plan must be active at time of purchase
- Start date: required; not in the past
- Payment must complete before membership activates
- Member must be authenticated (JWT validated)

8 Workflow

```
Member browses plans
  All active plans fetched and displayed with details

Member selects plan
  Start date selected (default: today)
  Order summary shown: plan name, price, validity, start/end dates

Member proceeds to payment
  Razorpay checkout initiated (test mode Phase 1)
  Member completes payment

Server receives Razorpay webhook (signed)
  Signature verified server-side
  Payment record updated: Paid
  Membership state: Pending → Active (auto-approval) or awaiting Admin approval
  WhatsApp: membership-activated sent

Member sees confirmation screen
  Membership card updated on Dashboard
```

9 Acceptance Criteria

- Plans list shows only active plans (Admin-controlled; deactivated plans not shown)
- Purchase creates Pending membership; activation only after verified payment
- Past date selection blocked in start date picker
- Payment failure shows retry option; membership remains Pending
- Activated membership visible immediately on member Dashboard card
- WhatsApp membership-activated sent after confirmed payment and activation

B2C-03 Bookings Module (Member-Facing)

1 Module Overview

Enables members to browse the live class schedule, book available sessions, cancel bookings within the allowed window, join the waitlist when sessions are full, and view their booking history.

2 Business Purpose

Replace WhatsApp and phone booking with self-service scheduling that is available 24x7, enforces eligibility rules automatically, and gives members instant confirmation.

3 Features

- **Live Schedule:** Filterable weekly/daily class timetable with real-time seat availability
- **Eligibility Indicator:** Sessions eligible under member's current plan highlighted; ineligible greyed out with reason
- **Book Session:** One-tap booking; immediate confirmation or waitlist placement
- **Cancellation:** Cancel within configured window (default 4 hours before session)
- **Waitlist:** Join waitlist when full; automatic WhatsApp notification when promoted
- **Booking History:** Upcoming bookings and past booking history with status

4 User Roles

Role	Access
Authenticated Member	Own bookings only; can view full schedule filtered by plan entitlement
Unauthenticated Visitor	Cannot book; schedule may be visible on public site

5 Functions

- getSchedule(tenantId, studioId, dateRange, membershipPlanId) — live schedule with eligibility flags
- getAvailability(sessionId) — current seats and waitlist count
- bookSession(memberId, sessionId) — atomic booking via Bookings service
- cancelBooking(bookingId, memberId) — cancellation with cutoff window check
- getBookingsForMember(memberId, tenantId) — upcoming and past bookings
- joinWaitlist(memberId, sessionId) — waitlist placement via Bookings service

6 Business Rules

- Final eligibility, capacity, duplicate, and booking limit validation always executed server-side at commit time
- Member cannot book a class type not included in their current active plan

- Member with Expired or Suspended membership sees all classes as ineligible (with reason)
- Self-cancellation allowed up to configured cutoff (default 4 hours); after cutoff, member must contact studio
- Waitlist is FIFO; member receives WhatsApp when promoted — no guarantee of seat until promotion confirmed
- Booking confirmation is the authoritative ERP result — not an optimistic client response

7 Validation Rules

- Member must be authenticated with active/valid session
- Member must have an Active membership at time of booking commit
- Session must be Scheduled (not Cancelled or Completed)
- Session must be in the future
- Class type must be in member's plan entitlement
- Member cannot have a duplicate booking for the same session

8 Workflow

```
Member opens schedule
  Weekly view rendered with real-time availability
  Sessions eligible under member's plan highlighted
  Full sessions show "Join Waitlist" option

Member selects session and taps Book
  API atomic booking call:
    Eligibility check: membership active and class type entitled
    Capacity check: seat available?
      YES → Booking Confirmed (Booked status)
      NO → Join waitlist? YES → Waitlisted
  Confirmation shown in app
  WhatsApp booking-confirmation sent

Member needs to cancel:
  Within 4h window → Cancel confirmed; seat released; waitlist promoted
  After 4h window → "Contact your studio to cancel"
  WhatsApp: booking-cancelled sent
  Promoted waitlist member notified automatically
```

9 Acceptance Criteria

- Schedule shows real-time availability; full sessions show Waitlist option
- Ineligible class type shown greyed out with reason "Not included in your plan"
- Booking response reflects authoritative ERP result within 1 second (NFR-PRF-01)
- Self-cancellation blocked after cutoff window (configurable; default 4 hours)
- Waitlist member automatically promoted and notified when seat opens
- Booking history shows Upcoming and Past tabs with correct status for each booking

B2C-04 Attendance Module (Member-Facing)

1 Module Overview

QR code self-check-in for class sessions and events, plus personal attendance history view. The member's proof-of-presence mechanism.

2 Business Purpose

Eliminate paper attendance lists and manual front-desk check-in for self-directed members. Give members a record of their own studio visits.

3 Features

- **QR Check-In:** Member scans session QR code from phone camera or B2C app scanner
- **Check-In Confirmation:** Immediate on-screen confirmation of successful check-in
- **Attendance History:** Chronological list of all sessions attended with date, class type, duration
- **Attendance Summary:** Sessions attended this week, this month, total (simple stats strip)

4 User Roles

Role	Access
Authenticated Member	Own attendance records only; QR check-in for own bookings
Unauthenticated Visitor	No access

5 Functions

- checkIn(memberId, sessionId, qrToken) — idempotent QR check-in via Attendance service
- getMemberAttendanceHistory(memberId, tenantId, filters) — full attendance history

6 Business Rules

- Member must have a confirmed Booking for the session to check in (BR-ATT-01 → member-facing)
- QR token is session-specific and time-limited (configurable window around session start)
- Duplicate check-in is idempotent — returns "Already checked in" confirmation, no error
- Member can view their own attendance history only — no other member's records visible
- Attendance check-in via B2C portal is equivalent to QR check-in method in audit record

7 Validation Rules

- QR token: validated server-side for session identity, expiry, and tampering
- Member JWT must match member_id derived from QR scan context
- Booking must be in Confirmed/Booked status for the session

- Session must be within configured check-in window (default: 30 min before to 30 min after start)

8 Workflow

```
Session QR code displayed (front desk screen, printed sheet, or B2C session detail)
Member opens B2C app → Navigate to Attendance → Scan QR
  Token transmitted to API
  API validates:
    Token is valid and not expired
    Token maps to a real session within check-in window
    Member has confirmed Booking for session
    No duplicate attendance record for this member+session
  If all valid: Attendance Record created (method: QR, status: Present)
  Confirmation shown in app: "Checked in to Morning Yoga — 7:00 AM"
  If already checked in: "You're already checked in for this session"
```

9 Acceptance Criteria

- QR token expiry validated server-side; expired token shows clear error
- Successful check-in shows confirmation message in app immediately
- Duplicate check-in shows "Already checked in" — no error, no duplicate record created
- Attendance history shows all sessions attended with correct dates, types, and statuses
- Member can only check in for sessions they have a confirmed Booking for (server-enforced)

B2C-05 Events Module (Member-Facing)

1 Module Overview

Allows members to browse upcoming events, register for eligible events (free or paid), view their event registrations, and receive event reminders.

2 Business Purpose

Drive event participation and revenue by giving members a frictionless, self-service event discovery and registration flow with instant payment confirmation.

3 Features

- **Event Discovery:** Browsable event list — upcoming workshops, studio events, special sessions
- **Event Filter:** By category, date, free/paid, available seats
- **Event Detail:** Description, instructor, date/time, duration, price, remaining seats, registration deadline
- **Registration Flow:** Register → Payment if required → Instant confirmation
- **My Registrations:** List of all past and upcoming event registrations with status
- **Cancellation:** Cancel registration if within allowed window (configurable)

4 User Roles

Role	Access
Authenticated Member	All events eligible for their audience type; self-registration
Unauthenticated Visitor	Public events only; must log in to complete registration

5 Functions

- getEvents(tenantId, studioId, filters, membershipPlanId) — event list filtered by audience type
- getEventDetail(eventId, tenantId) — full event info with seat count
- registerForEvent(memberId, eventId) — registration with capacity check and payment if required
- getMemberEventRegistrations(memberId, tenantId) — own registrations
- cancelEventRegistration(registrationId, memberId) — within allowed window

6 Business Rules

- Members-only events shown only to authenticated members (hidden from visitors)
- Paid event registration confirmed only after server-verified payment
- Capacity enforcement is atomic (same pattern as class booking)
- Registration window enforced: no registrations after configured cutoff before event
- Free events: registration confirmed immediately upon submission

- Paid events: registration pending until payment confirmed; seat held for configurable timeout (Phase 2)

7 Workflow

```
Member browses events (filtered by audience type)
Member selects event → Views full detail

Free event:
  Member taps Register
  Capacity check (atomic ERP)
  Registration confirmed immediately
  WhatsApp event-registration-confirmation sent

Paid event:
  Member taps Register → Proceeds to payment
  Razorpay checkout initiated
  Server verifies payment webhook
  Registration confirmed
  WhatsApp event-registration-confirmation and payment-received sent

Event day:
  Event reminder sent 24h, 6h, and 1h before event
  Member attends; attendance recorded via Attendance module
```

8 Acceptance Criteria

- Members-only events invisible to unauthenticated visitors (server-enforced)
- Paid event registration confirmed only after server-verified payment
- Event capacity enforced atomically (last-seat contention: exactly one confirmation)
- Registration cutoff window enforced; no registrations accepted after cutoff
- My Registrations shows upcoming and past registrations with correct status
- Event reminder WhatsApp sent at 24h, 6h, 1h before event start

B2C-06 Payments Module (Member-Facing)

1 Module Overview

Provides the member-facing checkout experience for membership purchases and event registrations, plus a personal payment history view.

2 Business Purpose

Enable members to pay online instantly without visiting the studio, with immediate confirmation and a clear payment record they can reference.

3 Features

- **Online Checkout:** Razorpay-powered checkout for memberships and events
- **Payment History:** All past payments with date, amount, purpose, status, reference number
- **Payment Status:** Paid, Pending, Failed with appropriate messaging and retry option
- **Receipt / Confirmation:** In-app confirmation on successful payment (WhatsApp confirmation also sent)

4 User Roles

Role	Access
Authenticated Member	Own payment checkout and history only
Unauthenticated Visitor	Cannot access any payment screen

5 Functions

- getPaymentsForMember(memberId, tenantId) — own payment history
- getCheckoutLink(memberId, purpose, metadata) — initiates Razorpay order and returns checkout link
- getPaymentStatus(paymentId, memberId) — polls or receives status of a specific payment

6 Business Rules

- All transactions server-side verified — client-side success callback never trusted
- Payment is always linked to a specific purpose: Membership assignment ID or Event registration ID
- Failed payment: member shown clear failure message with retry option; no membership or registration activated
- Member can view only their own payment records

7 Acceptance Criteria

- Checkout flow completes and returns control to app within reasonable time
- Failed payment: clear message and retry option shown; no accidental activation

- Payment history shows all transactions with correct status and amounts
- WhatsApp payment-received sent only after server-verified payment

B2C-07 Profile Module

1 Module Overview

Allows members to view and update their own profile information, manage authentication methods, set preferences, and view emergency contact details.

2 Business Purpose

Give members ownership of their own data. Reduce admin workload for routine updates like phone number changes.

3 Features

- **Profile View/Edit:** Name, email, date of birth, emergency contact, health notes (optional)
- **Authentication Management:** Phone number (primary login), Google account link/unlink
- **Phone OTP Login:** Login via OTP to verified phone number
- **Google Login:** OAuth2 login linked to member account
- **Account Linking:** Link Google account to existing phone-authenticated account
- **Preferences:** Communication preference (Phase 2 — WhatsApp opt preferences)
- **Account Deactivation Request:** Submit request to studio (handled by Admin)

4 User Roles

Role	Access
Authenticated Member	Own profile only
Unauthenticated Visitor	Login and registration flow only

5 Functions

- getMemberProfile(memberId, tenantId) — own profile data
- updateMemberProfile(memberId, changes) — permitted field updates with audit
- requestPhoneChange(memberId, newPhone) — sends OTP to new number; requires re-verification
- linkGoogleAccount(memberId, googleOAuthToken) — account linking
- unlinkGoogleAccount(memberId) — removes Google link (if primary auth is phone)

6 Business Rules

- Phone number is the primary login identifier — changes require OTP verification of new number
- Google account can be linked to an existing phone-authenticated account (unified identity)
- Member cannot unlink all authentication methods — must have at least one active auth
- Health notes are member-controlled and visible to Admin only when member explicitly consents

- Profile changes produce audit records (same audit pipeline as staff edits)
- Account deactivation must be requested through the studio (Admin-initiated archival)

7 Authentication Flows

```
Phone OTP Login:
  Member enters phone number
  OTP sent to phone
  Member enters OTP
  OTP validated server-side (single use, time-limited)
  JWT issued with memberId and tenantId claims

Google OAuth Login:
  Member taps "Login with Google"
  OAuth2 redirect to Google
  Callback returns ID token
  Server validates ID token with Google
  If linked account found → JWT issued
  If no linked account → Show link/register option

Account Linking:
  Member already logged in with phone
  Navigates to Profile → Auth Settings → Link Google Account
  Google OAuth flow → On success: Google account linked to member record
  Future: can log in with Google or phone OTP
```

8 Acceptance Criteria

- Phone change requires OTP verification of new number; old phone cannot be reused without re-verification
- Google login works after account linking; both auth methods produce same member identity
- Profile field updates produce audit records
- Member cannot accidentally unlink the only active authentication method
- OTP is single-use and expires within configured window (default 10 minutes)

B2C-08 Notifications Module (Member-Facing)

1 Module Overview

The member-facing notification experience. Phase 1 is entirely WhatsApp-based. Phase 2 adds in-app notification centre.

2 Phase 1 — WhatsApp Notifications (Received, Not Managed in App)

In Phase 1, members receive WhatsApp messages but do not manage notification preferences in the app. All WhatsApp messages sent to the member's registered phone number via approved templates.

MESSAGES RECEIVED BY MEMBERS

Trigger	WhatsApp Template
Booking confirmed	booking-confirmation
Booking cancelled (by staff)	booking-cancelled
Promoted from waitlist	seat-promotion-confirmation
Class reminder	class-reminder (12h and 1h before)
Membership activated	membership-activated
Renewal reminder	renewal-reminder (7d, 3d, 1d, expiry day)
Payment received	payment-received
Event registration	event-registration-confirmation
Event reminder	event-reminder (24h, 6h, 1h before)
Welcome (on member creation)	welcome-message (optional)

3 Phase 2 — In-App Notification Centre

- In-app notification bell with unread count
- Notification list: booking updates, membership alerts, event reminders, studio announcements
- Mark as read; clear all
- Notification preference management (opt-in/out per notification type)

4 Business Rules

- BR-NTF-04: Until production WhatsApp account exists, mock mode records messages without actual delivery
- BR-NTF-05: Only approved WhatsApp templates used; opt-outs honoured
- Member phone number is the WhatsApp delivery address — same as login phone
- Failed message does not reverse the business transaction that triggered it
- Members cannot suppress system-critical messages (e.g., booking cancellation by studio)

5 Acceptance Criteria

- Member receives WhatsApp for each trigger event listed above
- Message sent only after business transaction committed (not optimistic)
- Delivery status tracked and visible in admin Notification Log
- Mock mode in Phase 1: records would-send entry; admin can inspect

B2C-09 Recommendations Module (Member-Facing) — Phase 1 Narrow

1 Module Overview

AI-powered class suggestion feature that recommends sessions to the member based on their stated fitness goal, experience level, plan entitlement, and attendance history.

2 Business Purpose

Help members discover classes they will enjoy, increasing engagement, retention, and session utilisation rates.

3 Scope (Phase 1 — Narrow)

- Goal and level declared by member during onboarding or in profile
- AI reads: member's declared goal, experience level, plan entitlement, last 30 days attendance
- AI outputs: ordered shortlist of upcoming eligible sessions with reason for each suggestion
- Member clicks suggested session → standard booking flow

4 Features

- **Recommendations Strip:** 3–5 suggestions on Dashboard or dedicated Recommendations screen
- **Session Suggestion Card:** Class type, date/time, trainer, seats available, reason for suggestion ("Based on your goal: flexibility")
- **Book from Suggestion:** Direct link to standard booking flow; no separate booking via AI

5 User Roles

Role	Access
Authenticated Member	Own recommendations based on own profile and history
Unauthenticated Visitor	No access

6 Functions

- getRecommendations(memberId, tenantId) — AI recommendation tool via AI Platform
- AI tool calls: getMemberProfile, getMemberAttendanceSummary, getEligibleSchedule — all governed

7 Business Rules

- Recommendations are suggestions only — member always makes the final booking decision through standard flow
- AI only recommends sessions within the member's plan entitlement (no upscell pressure)

- Recommendations must be sessions that are scheduled, not cancelled, and in the future
- AI reads only summary data (attendance count by class type, declared goal) — not individual health notes
- Feature available only when tenant tier and feature flag permit (BR-AI-11)

8 AI Business Rules Applied

- BR-AI-01: AI acts only through governed tools — does not query database directly
- BR-AI-02: AI inherits member's scope — cannot recommend sessions from other studios
- BR-AI-04: AI does not ask for medical information — uses declared goal and level only
- BR-AI-05: AI reports actual available sessions — does not invent availability

9 Acceptance Criteria

- Recommendations appear within 3 seconds (NFR-PRF-02)
- All recommended sessions are eligible under member's current active plan (server-validated)
- All recommended sessions are upcoming (future), Scheduled (not Cancelled)
- Clicking recommendation navigates to standard booking flow for that session
- Feature absent when tenant tier or feature flag does not permit it
- AI uses only declared goal, level, and attendance summary — no medical data

SECTION 4

Complete Screen Inventory

4.1 B2B Portal Screens

Authentication and Onboarding

Screen ID	Screen Name	Route	Primary Role	Module	Priority
B2B-A-01	Login	`/login`	All Staff	Auth	P0
B2B-A-02	Forgot Password	`/forgot-password`	All Staff	Auth	P0
B2B-A-03	First Login Setup	`/setup`	Owner	Auth/Settings	P0
B2B-A-04	Accept Invitation	`/invite/:token`	New Staff	Auth/Staff	P0
B2B-A-05	MFA Setup	`/mfa-setup`	Owner, Admin	Auth	P2

Dashboard

Screen ID	Screen Name	Route	Primary Role	Module	Priority
B2B-D-01	Main Dashboard	`/dashboard`	All Staff	Dashboard	P0
B2B-D-02	Trainer Workspace	`/trainer`	Trainer	Dashboard	P0

CRM and Leads

Screen ID	Screen Name	Route	Primary Role	Module	Priority
B2B-C-01	Lead List	`/leads`	Owner, Admin, Front Desk	CRM	P0
B2B-C-02	Lead Detail	`/leads/:id`	Owner, Admin, Front Desk	CRM	P0
B2B-C-03	Create Lead Form	`/leads/new`	Owner, Admin, Front Desk	CRM	P0
B2B-C-04	Lead Pipeline View	`/leads/pipeline`	Owner, Admin	CRM	P0
B2B-C-05	Lead Source Report	`/leads/report`	Owner, Admin	CRM/Reports	P1
B2B-C-06	Convert Lead Modal	`in-modal`	Owner, Admin	CRM/Members	P0

Members

Screen ID	Screen Name	Route	Primary Role	Module	Priority
B2B-M-01	Member List	/members	All Staff	Members	P0
B2B-M-02	Member Detail	/members/:id	All Staff (scoped)	Members	P0
B2B-M-03	Member Detail - Memberships Tab	/members/:id/memberships	Owner, Admin	Members/ Memberships	P0
B2B-M-04	Member Detail -Bookings Tab	/members/:id/bookings	Owner, Admin, FD	Members/Bookings	P0
B2B-M-05	Member Detail -Attendance Tab	/members/:id/attendance	Owner, Admin, FD	Members/ Attendance	P0
B2B-M-06	Member Detail -Payments Tab	/members/:id/payments	Owner, Admin	Members/Payments	P0
B2B-M-07	Create Member Form	/members/new	Owner, Admin, FD	Members	P0
B2B-M-08	Edit Member Form	/members/:id/edit	Owner, Admin, FD	Members	P0
B2B-M-09	Duplicate Warning Modal	in-modal	Owner, Admin, FD	Members	P0

Memberships

Screen ID	Screen Name	Route	Primary Role	Module	Priority
B2B-MS-01	Plan List	/plans	Owner	Memberships	P0
B2B-MS-02	Create/Edit Plan Form	/plans/new, /plans/:id/edit	Owner	Memberships	P0
B2B-MS-03	Assign Membership Form	in-modal	Owner, Admin, FD	Memberships	P0
B2B-MS-04	Membership Approval Queue	/memberships/pending	Owner, Admin	Memberships	P0
B2B-MS-05	Renewal Pipeline	/memberships/expiring	Owner, Admin	Memberships	P0
B2B-MS-06	Membership State Action Modal	in-modal	Owner, Admin	Memberships	P0

Classes

Screen ID	Screen Name	Route	Primary Role	Module	Priority
B2B-CL-01	Schedule View (Weekly)	/classes	Owner, Admin, FD	Classes	P0

Screen ID	Screen Name	Route	Primary Role	Module	Priority
B2B-CL-02	Session Detail	/classes/:id	Owner, Admin, FD, Trainer	Classes	P0
B2B-CL-03	Create Session Form	/classes/new	Owner, Admin	Classes	P0
B2B-CL-04	Edit Session Form	/classes/:id/edit	Owner, Admin	Classes	P0
B2B-CL-05	Class Type Management	/class-types	Owner, Admin	Classes	P0
B2B-CL-06	Session Participant List	/classes/:id/participants	Owner, Admin, FD, Trainer	Classes	P0
B2B-CL-07	Cancel Session Modal	in-modal	Owner, Admin	Classes	P0

Bookings

Screen ID	Screen Name	Route	Primary Role	Module	Priority
B2B-BK-01	All Bookings List	/bookings	Owner, Admin	Bookings	P0
B2B-BK-02	Book on Behalf Modal	in-modal	Owner, Admin, FD	Bookings	P0
B2B-BK-03	Cancel Booking Modal	in-modal	Owner, Admin, FD	Bookings	P0
B2B-BK-04	Waitlist View (per session)	in-session-detail	Owner, Admin	Bookings	P0

Attendance

Screen ID	Screen Name	Route	Primary Role	Module	Priority
B2B-AT-01	Check-In Dashboard	/check-in	Front Desk	Attendance	P0
B2B-AT-02	Session Attendance Sheet	/classes/:id/attendance	Owner, Admin, FD, Trainer	Attendance	P0
B2B-AT-03	Member Attendance History	/members/:id/attendance	Owner, Admin, FD	Attendance	P0
B2B-AT-04	QR Code Display Screen	/classes/:id/qr	Front Desk	Attendance	P0
B2B-AT-05	Attendance Correction Modal	in-modal	Owner, Admin	Attendance	P0

Events

Screen ID	Screen Name	Route	Primary Role	Module	Priority
B2B-EV-01	Event List	/events	Owner, Admin, FD	Events	P0
B2B-EV-02	Event Detail	/events/:id	Owner, Admin, FD	Events	P0

Screen ID	Screen Name	Route	Primary Role	Module	Priority
B2B-EV-03	Create Event Form	/events/new	Owner, Admin	Events	P0
B2B-EV-04	Edit Event Form	/events/:id/edit	Owner, Admin	Events	P0
B2B-EV-05	Event Registrations List	/events/:id/registrations	Owner, Admin, FD	Events	P0
B2B-EV-06	Register Member for Event Modal	in-modal	Owner, Admin, FD	Events	P0
B2B-EV-07	Cancel Event Modal	in-modal	Owner, Admin	Events	P0

Payments

Screen ID	Screen Name	Route	Primary Role	Module	Priority
B2B-PY-01	Payment List	/payments	Owner, Admin	Payments	P0
B2B-PY-02	Record Manual Payment Modal	in-modal	Owner, Admin, FD	Payments	P0
B2B-PY-03	Payment Detail	/payments/:id	Owner, Admin	Payments	P0
B2B-PY-04	Payment Exception List	/payments/exceptions	Owner, Admin	Payments	P0

Reports

Screen ID	Screen Name	Route	Primary Role	Module	Priority
B2B-RP-01	Report Hub	/reports	Owner, Admin	Reports	P1
B2B-RP-02	Active Members Report	/reports/members	Owner, Admin	Reports	P1
B2B-RP-03	Revenue by Plan Report	/reports/revenue	Owner	Reports	P1
B2B-RP-04	Class Occupancy Report	/reports/occupancy	Owner, Admin	Reports	P1
B2B-RP-05	Attendance Trend Report	/reports/attendance	Owner, Admin	Reports	P1
B2B-RP-06	Renewal Pipeline Report	/reports/renewals	Owner, Admin	Reports	P1
B2B-RP-07	Lead Conversion Report	/reports/leads	Owner, Admin	Reports	P1

Staff Management

Screen ID	Screen Name	Route	Primary Role	Module	Priority
B2B-ST-01	Staff List	/staff	Owner, Admin	Staff	P0
B2B-ST-02	Staff Detail	/staff/:id	Owner, Admin	Staff	P0
B2B-ST-03	Invite Staff Form	/staff/invite	Owner, Admin	Staff	P0
B2B-ST-04	Edit Staff/Role Form	/staff/:id/edit	Owner	Staff	P0
B2B-ST-05	Trainer Profile	/staff/:id/trainer	Owner, Admin	Staff	P0

Screen ID	Screen Name	Route	Primary Role	Module	Priority
B2B-ST-06	Deactivate Staff Modal	in-modal	Owner	Staff	P0

Settings

Screen ID	Screen Name	Route	Primary Role	Module	Priority
B2B-SE-01	Settings Hub	/settings	Owner	Settings	P0
B2B-SE-02	Studio Profile Settings	/settings/studio	Owner	Settings	P0
B2B-SE-03	Plan Management (in Settings)	/settings/plans	Owner	Settings	P0
B2B-SE-04	Class Type Management (in Settings)	/settings/class-types	Owner, Admin	Settings	P0
B2B-SE-05	Notification Template Settings	/settings/notifications	Owner, Admin	Settings	P1
B2B-SE-06	Feature Flags Settings	/settings/features	Owner	Settings	P0
B2B-SE-07	Cancellation Window Settings	/settings/booking-rules	Owner	Settings	P0

Notifications and AI

Screen ID	Screen Name	Route	Primary Role	Module	Priority
B2B-NT-01	Notification Log	/notifications/log	Owner, Admin	Notifications	P1
B2B-AI-01	AI Assistant Panel	embedded in layout	Owner, Admin	AI Assistant	P1
B2B-AI-02	AI Action Confirmation Card	in-panel	Owner, Admin	AI Assistant	P1
B2B-AI-03	AI Audit Log	/ai/audit	Owner	AI Platform	P1

4.2 B2C Member Portal Screens

Authentication

Screen ID	Screen Name	Route	Module	Priority
B2C-A-01	Phone OTP Login	/login	Auth	P0
B2C-A-02	OTP Verification	/login/verify	Auth	P0
B2C-A-03	Google OAuth Callback	/auth/google/callback	Auth	P0
B2C-A-04	Account Link Flow	/account/link	Auth/Profile	P0
B2C-A-05	New Member Registration	/register	Auth/Members	P0

Dashboard and Home

Screen ID	Screen Name	Route	Module	Priority
B2C-D-01	Member Dashboard	/home	Dashboard	P0
B2C-D-02	Renewal CTA Screen	/renew	Membership	P0

Membership

Screen ID	Screen Name	Route	Module	Priority
B2C-MS-01	Plan Browsing	/plans	Membership	P0
B2C-MS-02	Plan Detail	/plans/:id	Membership	P0
B2C-MS-03	Purchase Checkout	/checkout/membership	Payments	P0
B2C-MS-04	Payment Confirmation	/checkout/confirmation	Payments	P0
B2C-MS-05	Membership History	/my-membership	Membership	P0
B2C-MS-06	Current Membership Detail	/my-membership/:id	Membership	P0

Bookings

Screen ID	Screen Name	Route	Module	Priority
B2C-BK-01	Class Schedule	/classes	Bookings	P0
B2C-BK-02	Session Detail	/classes/:id	Bookings	P0
B2C-BK-03	Book Confirmation Modal	in-modal	Bookings	P0
B2C-BK-04	My Bookings (Upcoming)	/my-bookings	Bookings	P0
B2C-BK-05	My Bookings (History)	/my-bookings/history	Bookings	P0
B2C-BK-06	Cancel Booking Modal	in-modal	Bookings	P0
B2C-BK-07	Waitlist Joined Confirmation	in-modal	Bookings	P0

Attendance

Screen ID	Screen Name	Route	Module	Priority
B2C-AT-01	QR Scanner /Check-In	/check-in	Attendance	P0
B2C-AT-02	Check-In Success Screen	/check-in/success	Attendance	P0
B2C-AT-03	My Attendance History	/my-attendance	Attendance	P0

Events

Screen ID	Screen Name	Route	Module	Priority
B2C-EV-01	Event Discovery	/events	Events	P0
B2C-EV-02	Event Detail	/events/:id	Events	P0
B2C-EV-03	Event Checkout	/checkout/event	Payments	P0
B2C-EV-04	My Event Registrations	/my-events	Events	P0

Payments and Profile

Screen ID	Screen Name	Route	Module	Priority
B2C-PY-01	Payment History	/my-payments	Payments	P0
B2C-PR-01	My Profile	/profile	Profile	P0
B2C-PR-02	Edit Profile Form	/profile/edit	Profile	P0
B2C-PR-03	Auth Settings	/profile/auth	Profile	P0
B2C-PR-04	Change Phone Flow	/profile/change-phone	Profile	P0

AI and Recommendations

Screen ID	Screen Name	Route	Module	Priority
B2C-AI-01	Recommendations Strip (on Dashboard)	embedded	Recommendations	P1
B2C-AI-02	Recommendations Screen	/recommendations	Recommendations	P1

4.3 Public-Facing Screens (Pre-Login)

Screen ID	Screen Name	Route	Module	Priority
PUB-01	Studio Landing Page	/	Public	P0
PUB-02	Public Class Schedule	/schedule	Public/Classes	P0
PUB-03	Public Event List	/events	Public/Events	P0
PUB-04	Enquiry/Lead Capture Form	/enquiry	CRM	P0
PUB-05	AI Enquiry Chat Widget	embedded on landing	CRM/AI	P1
PUB-06	Member Login/Register	/login	Auth	P0

4.4 Shared / System Screens

Screen ID	Screen Name	Context	Priority
SYS-01	404 Not Found	Any invalid route	P0
SYS-02	403 Forbidden	Unauthorised access attempt	P0
SYS-03	500 Error	System error fallback	P0
SYS-04	Maintenance Mode	Planned downtime notice	P0
SYS-05	Session Expired	JWT expired or revoked	P0

SECTION 5

Cross-Module Workflows

5.1 Member Onboarding Workflow

Trigger

Studio Admin or Front Desk creates a new member record (or member self-registers from B2C portal).

Actors

Staff (Admin/Front Desk) or Member (self-registration via B2C).

Flow

```
STEP 1: IDENTITY CREATION
  Staff enters: Name, Phone (required), Email (optional)
  System runs real-time duplicate check: phone and email within tenant
    Duplicate found → Modal: Link existing / Proceed as new / Cancel
    No duplicate → Continue

  Staff enters additional fields: Date of Birth, Emergency Contact, Notes
  Submit: API validates → ERP Member DocType created
  Member ID issued; Tenant scoping applied

STEP 2: AUTHENTICATION (B2C Login)
  OTP sent to member's phone (Phase 1: phone OTP)
  Member verifies phone → B2C portal access granted
  Optional: Link Google account

STEP 3: PLAN SELECTION AND PAYMENT
  Admin assigns plan OR member browses and selects plan in B2C
  Start date set (default: today)
  Order summary shown

  If online: Razorpay checkout → Server-side webhook → Payment: Paid
  If offline: Admin records manual payment → Payment: Paid

STEP 4: MEMBERSHIP ACTIVATION
  Auto-approval (if configured): Membership → Active immediately
  OR Manual approval: Admin approves → Active
  WhatsApp: membership-activated sent

STEP 5: OPERATIONAL READINESS
  Member appears in Active Members list on Dashboard
  Member can log into B2C portal
  Member can book eligible class sessions
  Renewal reminder automation scheduled (7d, 3d, 1d before expiry)
```

Business Rules Applied

- Duplicate detection: BR-MEM-01
- Membership states: BR-MSP-02
- Payment verification: BR-PAY-02
- Audit: all creation and state-change steps produce immutable audit records

Acceptance Criteria

- End-to-end member onboarding completes in under 5 minutes (operational KPI)
- Every step produces audit record
- Member cannot book before membership is Active (server-enforced)
- WhatsApp activation message received within 2 minutes of activation

5.2 Class Booking Workflow

Trigger

Member or Staff initiates a class booking.

Actors

Member (self-book via B2C), Admin/Front Desk (book on behalf via B2B).

Flow

```
STEP 1: SCHEDULE DISCOVERY
  Member browses schedule (B2C) or Admin selects session (B2B)
  Eligibility pre-filtering applied: ineligible sessions greyed out
  Live availability shown: seats available or "Join Waitlist"
STEP 2: BOOKING REQUEST
  Member or Staff selects session and confirms
  API receives booking request with: memberId, sessionId, JWT

STEP 3: ATOMIC VALIDATION AND COMMIT (ERP)
  ERP atomic call with row lock on session:
    Validate: membership active and class type entitled
    Validate: booking limit not exceeded
    Validate: no duplicate booking for same session
    Validate: seat available?

  Case A — Seat Available:
    Booking record created (status: Booked)
    Session booking count incremented atomically
    Audit record created
    Outbox event: booking.confirmed
    Response: "Booking Confirmed"

  Case B — No Seat, Waitlist Enabled:
    Booking record created (status: Waitlisted) with queue position
    Outbox event: booking.waitlisted
```

```
    Response: "Added to Waitlist — Position N"

  Case C — No Seat, No Waitlist:
    No booking record created
    Response: "Session is Full"

  Case D — Ineligible:
    No booking record created
    Response: "Your plan does not include this class type" (or specific reason)

STEP 4: NOTIFICATION
  n8n booking-confirmation.json triggered by outbox event
  WhatsApp booking-confirmation sent to member

STEP 5: REMINDER SEQUENCE
  n8n class-reminders.json: scheduled to fire 12h and 1h before session
  WhatsApp class-reminder sent at each trigger

STEP 6: CHECK-IN (SESSION DAY)
  QR code generated for session
  Member scans QR → Attendance recorded
  OR Front Desk manually checks in member
  OR Trainer marks attendance post-session
```

Business Rules Applied

- Eligibility: BR-BKG-01, BR-MSP-05
- Atomic capacity: NFR-PRF-03
- Waitlist: BR-BKG-04
- Notifications: BR-NTF-02, BR-NTF-03

5.3 Membership Renewal Workflow

Trigger

Automated renewal reminder OR member/admin initiates manual renewal.

Actors

n8n scheduler (automated reminder), Member (self-serve via B2C), Admin (manual initiation).

Flow

```
AUTOMATED REMINDER SEQUENCE:
  Memberships service queries: memberships expiring in exactly 7 days
  n8n renewal-reminders.json triggered (daily schedule)
  WhatsApp renewal-reminder-7d sent to each expiring member

  Repeated at: 3 days before, 1 day before, expiry day
MEMBER-INITIATED RENEWAL (B2C):
  Member receives renewal CTA on Dashboard (expires ≤7 days)
  Member selects plan (same or different)
  Payment flow (same as new membership purchase)
  New Membership record created, linked to previous
  WhatsApp: membership-activated for new period
```

```
ADMIN-INITIATED RENEWAL (B2B):
  Admin navigates to Member → Memberships tab → Renew
  Selects plan; records payment (manual or initiates Razorpay)
  Payment confirmed → Membership activated
  WhatsApp: membership-activated

EXPIRY PROCESSING:
  On expiry day: WhatsApp membership-expired sent
  Membership state: Active → Expired
  Member blocked from booking restricted classes (server-enforced)
  Member still has B2C portal access to view history and renew
```

Business Rules Applied

- Manual renewal Phase 1 only: BR-MSP-04
- Payment independent of approval: BR-PAY-03
- Notifications: BR-NTF-03 (idempotency — one message per event per member)

5.4 Lead-to-Member Conversion Workflow

Trigger

Lead in CRM with status "Trial Done" or similar; Admin initiates conversion.

Actors

Admin (conversion action), system (source preservation), WhatsApp (confirmation).

Flow

```
PRE-CONDITION: Lead exists in ERP with status New/Contacted/Trial Done

STEP 1: CONVERSION INITIATION
  Admin opens Lead Detail → Click "Convert to Member"
  System checks: phone not already a member in this tenant (duplicate check)
  If duplicate found: confirmation modal with options (link/proceed/cancel)

STEP 2: MEMBER CREATION
  Member DocType created with:
    All fields from Lead (name, phone, email)
    Source: lead.source (immutable — preserved from original lead)
    Campaign: lead.campaign_tag (preserved)
    lead_id: linked to original Lead
  Audit record: actor, timestamp, from-lead reference

STEP 3: LEAD STATUS UPDATE
  Lead status → Converted
  Lead linked to newly created Member record
  Conversion timestamp recorded

STEP 4: MEMBERSHIP ASSIGNMENT (OPTIONAL — IMMEDIATE)
  Admin may immediately assign a membership plan
  Standard membership assignment flow begins
```

```
STEP 5: WELCOME COMMUNICATIONS
  Optional: WhatsApp welcome-message sent to new member
  Member can now log in to B2C portal
STEP 6: REPORTING
  Lead Conversion Report updated
  Source attribution preserved on Member record for long-term tracking
```

Business Rules Applied

- BR-CRM-02: Lead becomes Member only through explicit conversion event
- BR-CRM-01: Source always set at lead creation; preserved on conversion
- Source and campaign preserved on Member record permanently

5.5 Event Registration Workflow

Trigger

Member or Visitor discovers event and initiates registration.

Actors

Member/Visitor (self-registration), Admin/FD (staff-initiated registration).

Flow

```
STEP 1: EVENT DISCOVERY
  Member/Visitor browses event list (audience-type filtered)
  Selects event → Views full detail (capacity, price, registration deadline)

STEP 2: REGISTRATION REQUEST
  Member/Visitor taps Register
  System validates:
    Audience type eligibility (members-only vs. open)
    Registration window (before cutoff)
    Seat availability (atomic ERP)

FREE EVENT:
  Registration confirmed immediately
  Event Registration DocType created (status: Confirmed)
  Outbox: event.registration.confirmed
  WhatsApp: event-registration-confirmation sent

PAID EVENT:
  Razorpay checkout initiated
  Payment record created (status: Pending)
  Seat tentatively reserved (Phase 1: no hard hold — first confirmed payment wins)
  Member completes payment on Razorpay
  Server webhook verifies payment
  Payment: Paid
  Registration: Confirmed
  Outbox: event.registration.confirmed and payment.confirmed
  WhatsApp: event-registration-confirmation and payment-received sent

STEP 3: REMINDER SEQUENCE
  n8n event-reminder scheduled to fire at 24h, 6h, 1h before event
```

```
STEP 4: ATTENDANCE (EVENT DAY)
  Admin or assigned Trainer opens Event Attendance
  Marks each registrant Present or Absent (or QR check-in if configured)
```

Business Rules Applied

- BR-EVT-02, BR-EVT-04: audience type, paid confirmation rules
- Atomic capacity: same as class booking
- BR-NTF-03: idempotent messages (one per event + recipient)

5.6 AI Assistant Create Member Workflow

Trigger

Admin types a Create Member intent in the AI Assistant panel.

Actors

Admin (typing intent), AI Orchestrator (intent parsing), Admin (confirming), System (executing).

Flow

```
STEP 1: INTENT RECOGNITION
  Admin types: "Add new member Priya Sharma, phone 9876543210, Premium plan starting Monday"
  Message sent to AI Orchestrator with: user_id, tenant_id, role, session context
  Orchestrator builds system prompt with permitted tools for this user's role

STEP 2: FIELD EXTRACTION
  LLM calls extractCreateMemberFields tool
  Extracts:
    name: "Priya Sharma"
    phone: "9876543210"
    plan: "Premium" (resolved to plan_id by tool)
    start_date: next Monday's date (relative date resolved to ISO date)

STEP 3: MISSING FIELD RESOLUTION
  LLM checks mandatory fields: name (✓), phone (✓), plan (✓), start_date (✓)
  If any mandatory field missing: AI asks one question at a time
    "What plan should I assign to this member?"
    "What should the membership start date be?"

STEP 4: CONFIRMATION CARD
  AI renders confirmation card in chat:
    ┌──────────────────────────────────────┐
    | CREATE MEMBER SUMMARY                |
    | Name: Priya Sharma                   |
    | Phone: 9876543210                    |
    | Plan: Premium (₹2,999/month)         |
    | Start Date: Monday, 6 Oct 2026       |
    |                                      |
    | [Confirm]  [Cancel]                  |
    └──────────────────────────────────────┘
  Arguments stored in Pending Action store (PostgreSQL) at this point
```

```
STEP 5: ADMIN CONFIRMATION
  Admin clicks Confirm (or Cancel)
  If Cancel: Pending action expired; session ends
  If Confirm:
    Arguments retrieved from Pending Action store (fixed at Step 4 time — no drift)
    createMember capability called via Members service
    Members service calls ERP (same as B2B form submission)

STEP 6: RESULT REPORTING
  ERP returns result:
    Success: Member ID and name
    Error: Validation error or duplicate message

  AI reports actual result truthfully:
    Success: "Member Priya Sharma created successfully. Member ID: MEM-00123"
    Error: "I could not create the member. The phone number 9876543210 already belongs to an existing
member."

  AI Audit record created (immutable):
    user: Admin ID
    tool: createMember
    arguments: (logged)
    result: Success/Error
    timestamp: ISO timestamp
    pending_action_id: reference

STEP 7: DUPLICATE PROTECTION
  If ERP returns duplicate-phone error:
    AI reports error text verbatim
    AI does not retry automatically
    AI suggests: "Would you like me to look up the existing member with that phone number?"
```

Business Rules Applied

- BR-AI-01 through BR-AI-07
- Create Member via AI calls same createMember function as B2B form
- Arguments fixed at confirmation time — admin cannot be shown different data from what was confirmed

5.7 Payment Failure Recovery Workflow

Trigger

Razorpay payment attempt fails or webhook is not received within expected window.

Flow

```
ATTEMPT:
  Member initiates payment → Razorpay order created → Payment: Pending

FAILURE CASE A — Payment explicitly failed:
  Razorpay fires webhook with failure status
  API receives webhook → Verifies signature
  Payment status: Failed
  Membership remains in Pending state
  Admin exception list updated
  Member sees failure message in checkout screen with retry option
```

```
FAILURE CASE B — Webhook not received (delayed or lost):
  Payment remains in Pending state
  Member sees "Payment processing..." message
  Admin exception list shows unresolved pending payments older than threshold
  Admin can manually verify with Razorpay dashboard and record manual payment if needed

RETRY FLOW:
  Member clicks Retry → New Razorpay order created
  Original order payment record remains (Failed)
  New payment record created (Pending)
  If new payment succeeds → Membership activated via this new payment record
  No duplicate Membership activation (idempotency check: membership in Pending state receives one
activation)

NEVER OCCURS:
  System does not activate membership based on client-side success callback alone
  System does not assume payment is complete because checkout was initiated
```

5.8 Notification Delivery Workflow

Trigger

Any business event that produces an outbox event (booking, payment, membership state change, etc.)

Flow

```
Business transaction commits in ERP (or API)
  Outbox record written atomically with transaction:
    event_id (UUID), event_type, tenant_id, payload, created_at

Outbox Dispatcher (background service):
  Polls outbox for unprocessed events
  For each event:
    Signs payload with HMAC-SHA256
    Fires signed HTTP webhook to n8n webhook URL
    Marks outbox record as dispatched

n8n receives webhook:
  Validates HMAC signature
  Routes to correct workflow by event_type
  Workflow:
    Fetches recipient phone from Members service (using member_id from payload)
    Selects WhatsApp template for event type
    Fetches any template variables (e.g., member name, class name, expiry date)

    Mock mode (Phase 1):
      Writes would-send record to Notification Log (status: mock)
      Does NOT call WhatsApp Cloud API

    Live mode (Phase 2+):
      Calls WhatsApp Cloud API with template + variables
      WhatsApp API returns delivery status
      Notification Log updated with delivery status

  Idempotency check:
    If event_id already exists in Notification Log → Skip (already processed)

Failure handling:
  n8n retry with exponential backoff (3 attempts)
```

```
If all retries fail: failure-alert.json triggered
Admin notification panel updated with failure
Business transaction not reversed
```

SECTION 6

AI Module Specifications

6.1 AI Platform Architecture

Core Design Principles

The AI platform is governed, tool-based, and agent-ready. Every AI interaction flows through the AI Orchestrator — a single entry point that enforces security, scoping, tool filtering, and audit before any LLM call is made.

```
REQUEST FLOW:

User input
    |
    ▼
AI Orchestrator
    ├── Validates JWT (user identity, tenant, role)
    ├── Checks tenant tier and feature flags (BR-AI-11)
    ├── Filters Tool Registry to permitted tools for this user+role+tenant
    ├── Builds system prompt (scoped tool definitions + context)
    ├── Calls LLM Gateway (model-agnostic adapter)
    |
    ▼
LLM (commercial provider — Gemini, GPT-4o, or configured model)
    ├── Returns intent + tool selection + arguments
    |
    ▼
AI Orchestrator (tool dispatch)
    ├── Validates tool call against permitted registry
    ├── Calls governed domain service (Members, Classes, Memberships, Reports, etc.)
    ├── Service returns structured data
    |
    ▼
LLM (second pass — summarisation or confirmation)
    ├── Summarises ONLY from returned tool data
    ├── Never synthesises data not returned by tools
    |
    ▼
Response to user interface
    |
    ▼
AI Audit Log (immutable record — always)
```

AI Platform Components

Component	Purpose	Technology
AI Orchestrator	Central entry point; session management, tool filtering, audit	NestJS service

Component	Purpose	Technology
LLM Gateway	Model-agnostic adapter; swappable provider per environment	NestJS adapter (Gemini/ GPT-4o)
Tool Registry	Filtered list of permitted tools per user role and tenant tier	PostgreSQL + in-memory config
Pending Action Store	Holds write-action arguments between extraction and confirmation	PostgreSQL with TTL expiry
AI Audit Log	Immutable per-call record: user, tool, args, result, timestamp	PostgreSQL (append-only)
Context Engine	Conversation session state (ephemeral per session)	In-memory (NestJS)
Evaluation Harness	CI test suite for accuracy targets and zero-false-success	Jest + curated test set
Guardrails Module	Scope refusal, PII minimisation, health data prohibition, output grounding	Prompt + validation layer

Governed AI Business Rules (All AI Modules)

Rule	Description
BR-AI-01	AI acts only through governed capabilities; never reads or writes database directly
BR-AI-02	AI inherits signed-in user's permissions and tenant scope; cannot exceed them
BR-AI-03	Transactional actions require explicit user confirmation before execution
BR-AI-04	Assistant never guesses missing mandatory business data; asks for it
BR-AI-05	Assistant reports actual result from platform; never claims success not received
BR-AI-06	Every AI action logged immutably with initiator, capability, arguments, result, timestamp
BR-AI-07	Conversation context is temporary; becomes business data only through governed capability
BR-AI-08	AI never provides medical, legal, or financial advice; scope refusal enforced
BR-AI-09	AI cannot modify data presented in confirmation summary; only admin override by cancelling
BR-AI-10	PII minimisation: AI receives only necessary data from tool results; does not persist member data in AI context
BR-AI-11	AI features gated by subscription tier and feature flag per tenant

6.2 AI-01: Admin Assistant (Phase 1 MVP)

Overview

A natural-language operational assistant embedded in the B2B Studio Management Portal. Enables Admin and Owner to query operational data and execute the Create Member action conversationally.

Permitted Tools (Phase 1)

Tool ID	Tool Name	Action Type	Governed Service
T-01	`getExpiringMemberships`	Read	Memberships service
T-02	`getLowAttendanceClasses`	Read	Attendance/Reports service
T-03	`extractCreateMemberFields`	Internal	AI Orchestrator
T-04	`createMember`	Write (confirmation required)	Members service
T-05	`getMemberCount`	Read	Reports service
T-06	`getDashboardSummary`	Read	Reports service

Query Scenarios (Phase 1)

Query	Tool Used	Expected Output
"Which memberships expire this week?"	getExpiringMemberships (daysAhead=7)	List: member name, plan, expiry date
"Which classes have low attendance?"	getLowAttendanceClasses (threshold=50%)	Sessions with attendance below threshold
"How many active members do we have?"	getMemberCount (status=Active)	Count with optional breakdown by plan
"Create a new member Rahul Kumar, phone 9876543210, Premium plan starting Monday"	extractCreateMemberFields → createMember (with confirmation)	Confirmation card → execution

Out-of-Scope Responses (Phase 1)

The assistant must respond to out-of-scope questions with a scope refusal. Examples:

- "What is the best yoga pose for back pain?" → "I can help with studio operations like membership queries, attendance, and member management. I cannot provide fitness or health advice."
- "Can you send a WhatsApp to all members?" → "I can help you query member data. Bulk messaging is managed through the Notifications settings."
- "What is our tax liability?" → "I can help with operational reports. For financial tax questions, please consult your accountant."

Guardrails

- Output grounding: LLM summarises only from tool-returned data; never synthesises
- PII minimisation: member names and phones returned by tools only; not retained in conversation context beyond session
- No health/medical advice: hard-wired scope refusal
- Scope ceiling: reads and Create Member only in Phase 1; no bulk actions, no financial modifications

Accuracy Targets (from PRD)

- 85%+ of curated test queries answered correctly without escalation
- 95%+ of confirmed Create Member actions succeed on first confirmation
- 0 false-success reports

Acceptance Criteria

- All listed queries produce accurate answers matching corresponding Reports/Dashboard data
- Scope refusal fires correctly for all defined out-of-scope query types
- Confirmation card shown before any Create Member execution
- Create Member error reported truthfully — no false success
- AI Audit record created for every call
- Feature unavailable when tier flag is off (server-enforced)

6.3 AI-02: Enquiry Assistant (Phase 1 Narrow)

Overview

A public-facing first-response chatbot on the studio's landing page or enquiry channel. Answers common prospect questions and captures leads by initiating the Lead creation flow.

Permitted Tools

Tool ID	Tool Name	Action Type	Governed Service
T-07	`getPublicEventList`	Read	Events service (public audience only)
T-08	`getPublicClassSchedule`	Read	Classes service (public schedule only)
T-09	`getActivePlansPublic`	Read	Memberships service (active plans only)
T-10	`captureLead`	Write (no confirmation required — lead only, not member)	CRM service

Query Scenarios

Query	Tool Used	Expected Output
"What classes do you offer?"	getPublicClassSchedule	Active class types and schedule overview
"How much does membership cost?"	getActivePlansPublic	Plan names, prices, validity
"Do you have any upcoming workshops?"	getPublicEventList	Upcoming public events

Query	Tool Used	Expected Output
"I want to join /interested"	captureLead	Lead captured; WhatsApp confirmation to prospect

Lead Capture Flow

```
Visitor expresses interest in joining
AI collects: name (required), phone (required), email (optional), interest area
AI calls captureLead with collected fields
Lead created in ERP (same as manual Lead Capture form)
AI responds: "Thanks! A member of our team will reach out shortly."
n8n: WhatsApp lead-registration-confirmation sent to visitor's phone
```

Guardrails

- AI can create Leads only — never creates Members directly
- AI does not access any member-private data (attendance, payments, bookings)
- Public schedule data only — no member-restricted information returned
- Consent: lead capture includes implicit consent notice per template requirement

Acceptance Criteria

- Responds to all listed query scenarios accurately
- Lead creation calls same createLead function as B2B manual form
- Lead created in ERP with source = AI_ENQUIRY
- Scope refusal for questions outside defined capability (e.g., "Can you cancel my membership?")

6.4 AI-03: Booking Assistant (Phase 1 Narrow)

Overview

Helps members find suitable class sessions in natural language. Presents a shortlist of eligible, available sessions. Member confirms booking through the standard booking flow — AI does not book.

Permitted Tools

Tool ID	Tool Name	Action Type	Governed Service
T-11	`getEligibleSchedule`	Read	Classes + Memberships service (eligibility-filtered)
T-12	`getAvailability`	Read	Bookings service
T-13	`getMemberPlanEntitlement`	Read	Memberships service

Query Scenarios

Query	Tool Used	Expected Output
"Are there any yoga classes next week?"	getEligibleSchedule (type=Yoga, week=next)	Sessions eligible for member's plan
"I want something Tuesday morning"	getEligibleSchedule (day=Tuesday, time=morning)	Eligible sessions on Tuesday AM
"Is the 7am Pilates full?"	getAvailability (sessionId)	Seats available /waitlist count

Key Constraints

- AI returns a shortlist — member navigates to standard booking screen to complete booking
- AI never calls bookSession — booking is always a deliberate member action
- Recommendations limited to member's plan entitlement (no suggestions for ineligible classes)

Acceptance Criteria

- Shortlist contains only eligible, available, future sessions
- AI does not complete booking on member's behalf
- Availability shown is real-time (from live Bookings service)

6.5 AI-04: Recommendation Assistant (Phase 1 Narrow)

Overview

Proactively suggests class sessions based on member's declared goal, experience level, and recent attendance history.

Permitted Tools

Tool ID	Tool Name	Action Type	Governed Service
T-14	`getMemberGoalAndLevel`	Read	Members service (profile summary only)
T-15	`getMemberAttendanceSummary`	Read	Attendance service (class type counts, no individual session detail)
T-16	`getEligibleSessionsForRecommendation`	Read	Classes + Memberships service

Recommendation Logic

```
1. Read member's declared goal (e.g., flexibility, strength, weight loss) and level (beginner/intermediate/advanced)
2. Read attendance summary: sessions attended per class type in last 30 days
3. Fetch eligible upcoming sessions (next 7 days) filtered by plan entitlement
4. Rank sessions by:
   a. Alignment with declared goal (class types associated with goal)
   b. Variety: class types not recently attended weighted higher
   c. Availability: sessions with seats weighted higher
5. Return top 3-5 sessions with reason per suggestion
```

Guardrails

- Goal and level are declared by member — AI never infers or assigns health status
- AI reads attendance counts by class type only — not individual session records
- AI does not read health notes, payment data, or emergency contact
- AI only suggests sessions within member's plan entitlement
- Feature available only when tier flag permits

Acceptance Criteria

- Recommendations include only eligible, future, Scheduled sessions
- Reason for each recommendation is displayed
- Member profile health notes are never accessed
- Recommendations respect plan entitlement (no upsell pressure)

6.6 AI-05: Feedback Assistant (Phase 1 Narrow)

Overview

Processes post-class free-text comments submitted by members. Classifies sentiment, identifies topics, and generates a session summary for Admin and Trainer review.

Permitted Tools

Tool ID	Tool Name	Action Type	Governed Service
T-17	`classifyFeedback`	Internal	AI Orchestrator + LLM
T-18	`storeFeedbackClassification`	Write	Attendance/Feedback service
T-19	`getFeedbackSummary`	Read	Feedback service (for Admin/Trainer view)

Feedback Processing Flow

```
Member submits post-class comment (text field; optional; free text)
AI classifyFeedback tool processes comment:
  Sentiment: Positive / Neutral / Negative
  Topics: Trainer, Class Content, Music, Timing, Duration, Facility
  Theme: what the feedback is about in one sentence

Classification stored against session and class type (not shared publicly)
Admin/Trainer views Feedback Summary per session or per class type over period
```

Guardrails

- Feedback is always optional — members never compelled to provide it
- Raw text stored with attribution (member_id); classification stored separately
- AI does not provide individual member feedback directly to trainers — only aggregate session summary
- AI never uses feedback comments to make recommendations to other members

Acceptance Criteria

- Sentiment classification accuracy: measurable against labelled test set in evaluation harness
- Classification stored correctly against session
- Admin session feedback summary shows sentiment breakdown and top topics
- Raw feedback accessible only to Owner and Admin — not to Trainer or other members

SECTION 7

AI Agent Specifications (Phase 2+)

7.1 Agent Architecture (Phase 2 Foundation)

AI Agents differ from Phase 1 Assistants in autonomy and scope. Phase 2 Agents operate continuously or on schedule, can chain multiple tool calls, maintain short-term memory within a task, and always surface proposals for human approval before taking write actions.

Agent Safety Model

Control	Implementation
Human-in-the-loop	Write actions proposed, displayed, require explicit Admin approval
Scope boundary	Agent tool set is whitelisted per agent type; cannot exceed
Audit	All agent tool calls logged identically to human AI tool calls
No autonomous messaging	Agents never send WhatsApp or emails without Admin approval
Reversibility preference	Agents prefer reversible recommendations over irreversible actions
Rate limiting	Agent calls rate-limited per tenant to prevent runaway tool calls

7.2 CRM Agent (Phase 2)

Overview

An agent that monitors the CRM lead pipeline, scores leads by engagement, drafts follow-up messages for Admin review, and flags leads that have gone cold without follow-up.

Agent Tools

Tool	Action Type
`getLeadsWithStatus`	Read
`scoreLead (last contact, days since created, response count)`	Read + internal compute
`draftFollowUpMessage`	Internal (LLM draft)
`flagColdLead`	Write (Admin approval required)
`proposeStatusChange`	Write (Admin approval required)

Trigger

Daily or on-demand Admin trigger.

Human Approval Points

- Draft follow-up message: Admin reviews and sends manually (or approves for AI-assisted send — Phase 3)
- Status change proposal: Admin approves or rejects
- Cold lead flagging: surfaced in CRM as a suggested action

Acceptance Criteria

- Agent never sends messages autonomously
- All proposals displayed as actionable admin suggestions with one-click approve/reject
- All agent tool calls produce AI Audit records

7.3 Operations Agent (Phase 2)

Overview

Monitors class capacity and attendance patterns. Alerts Admin to underperforming sessions, suggests schedule adjustments, and identifies peak demand opportunities.

Agent Tools

Tool	Action Type
`getSessionOccupancyHistory`	Read
`getAttendanceTrendByClassType`	Read
`detectLowOccupancySessions`	Read + internal compute
`detectOversubscribedPatterns`	Read + internal compute
`proposeSessionCancellation`	Write (Admin approval required)
`proposeCapacityIncrease`	Write (Admin approval required)
`proposeScheduleChange`	Read (suggestion only)

Alert Examples

- "Morning Yoga sessions on Thursdays have averaged 22% occupancy over the last 4 weeks. Consider reducing capacity or rescheduling."
- "Saturday 10 AM Zumba has been fully booked with a waitlist of 8+ for 6 consecutive weeks. Consider adding a second session."

Human Approval Points

- Session cancellation: Admin must confirm
- Capacity increase: Admin must confirm; ERP enforces
- Schedule change: Advisory only — Admin action required in Classes module

7.4 Marketing Agent (Phase 3)

Overview

Drafts promotional campaigns, identifies audience segments for campaigns, and proposes WhatsApp or email campaigns for Admin approval and sending.

Key Constraint

Marketing Agent NEVER sends messages. All campaign drafts require Admin review and explicit approval. Sending is a human action.

Agent Tools (Proposed)

Tool	Action Type
`segmentMembersForCampaign`	Read
`draftCampaignMessage`	Internal (LLM draft)
`proposeCampaign`	Write (Admin approval required to schedule)
`getEventRegistrationList`	Read

Acceptance Criteria

- Campaign drafts marked as "Draft — Admin Review Required"
- No message sent without Admin explicit approval action
- Audience segment never includes members who have opted out

7.5 Growth Optimization Agent (Phase 3)

Overview

Advisory-only agent. Analyses revenue, occupancy, and membership trends to suggest pricing adjustments, class additions, and schedule optimisations.

Key Constraint

Growth Optimization Agent produces recommendations only. It never changes prices, plans, or schedules. All recommendations are displayed as advisory insights for Owner review.

Agent Tools (Proposed)

Tool	Action Type
`getRevenueTrend`	Read
`getOccupancyTrend`	Read
`getMemberRetentionRate`	Read
`compareRevenueByPlan`	Read
`generateGrowthRecommendations`	Internal (LLM analysis)

Output Format

Structured recommendation cards displayed in Owner's Reports screen:

- Observation: "Premium plan accounts for 72% of revenue but only 41% of active members."
- Recommendation: "Consider reviewing the Basic plan pricing or adding a benefit to increase Premium plan conversion."
- Confidence: Medium (based on N=83 data points)

7.6 Studio Insights Agent (Phase 3)

Overview

Proactive business intelligence agent. Generates weekly or daily studio health summaries, detects anomalies (sudden attendance drop, payment failure spike), and forecasts demand.

Trigger

Scheduled (daily/weekly) or event-triggered (anomaly threshold breached).

Output Channels

- Owner Dashboard Insights panel (B2B)
- Optional: WhatsApp summary to Owner (with explicit opt-in)

Key Constraint

All insights are read-only and advisory. No write actions.

SECTION 8

Automation Specifications (n8n Workflows)

8.1 Automation Architecture

n8n Design Principles

- n8n is the orchestration layer only — it contains no business logic
- Business decisions are made by the API (NestJS) — n8n acts on events only
- All workflows triggered by signed outbox events from the API
- Idempotency enforced by event_id: same event ID never triggers duplicate sends
- Mock/live mode toggled by environment variable — no code change required for production
- Every send attempt produces a Notification Log record (recipient, template, status, timestamp)

n8n Infrastructure

Component	Detail
Hosting	Self-hosted n8n on dedicated container
Authentication	HMAC-SHA256 signature verification on all inbound webhooks
Retry Policy	3 attempts with exponential backoff per workflow
Failure Routing	persistent failures → failure-alert.json → Admin notification panel
Logging	All workflow executions logged; inspectable in n8n UI by Platform Admin
Environment Variables	WHATSAPP_PHONE_NUMBER_ID, WHATSAPP_ACCESS_TOKEN, MOCK_MODE, API_SECRET
Shared Sub-workflow	send-message.json called by all notification workflows

Outbox Event Contract

Every outbox event received by n8n has this structure:

```
{
  "event_id": "uuid-v4",
  "event_type": "booking.confirmed | membership.activated | payment.confirmed | ...",
  "tenant_id": "tenant-uuid",
  "studio_id": "studio-uuid",
  "member_id": "member-uuid-or-null",
  "payload": { "...event-specific data..." },
  "timestamp": "ISO-8601"
}
```

8.2 Workflow 01: Lead Confirmation

FILE	lead-confirmation.json
TRIGGER	Outbox event — lead.created
PURPOSE	Confirm to the prospect that their enquiry was received.

Step	Action
1	Receive signed webhook (event_type: lead.created)
2	Verify HMAC signature
3	Extract: lead_id, name, phone from payload
4	Check idempotency: event_id not in Notification Log
5	Fetch template variables: lead name, studio name
6	Call send-message.json sub-workflow with phone, template=lead-registration-confirmation, variables
7	Record result in Notification Log

WHATSAPP TEMPLATE	lead-registration-confirmation
VARIABLES	lead_name, studio_name
TIMING	Immediately on lead creation

8.3 Workflow 02: Booking Confirmation

FILE	booking-confirmation.json
TRIGGER	Outbox event — booking.confirmed OR booking.waitlisted OR booking.promoted
PURPOSE	Confirm booking or waitlist placement; notify on waitlist promotion.

Trigger Event	Template	Variables
booking.confirmed	`booking-confirmation`	member_name, class_name, date, time, trainer_name

Trigger Event	Template	Variables
booking.waitlisted	`waitlist-confirmation`	member_name, class_name, date, time, queue_position
booking.promoted	`seat-promotion-confirmation`	member_name, class_name, date, time

FLOW

```
Receive signed webhook
Verify HMAC signature
Idempotency check (event_id)
Route to correct template by event_type
Fetch recipient phone from Members service (using member_id)
Fetch session details from Classes service (class_name, date, time, trainer)
Call send-message.json with template and variables
Record in Notification Log
```

TIMING	Immediately on booking commit

8.4 Workflow 03: Class Reminder

FILE	class-reminders.json
TRIGGER	n8n scheduled job (runs every 30 minutes to catch upcoming sessions)
PURPOSE	Send reminder WhatsApp 12 hours and 1 hour before each class session.

FLOW

```
Scheduled trigger fires (every 30 minutes)
Call API: getSessionsWithBookings(starts_in_hours=[12, 1])
  API returns: sessions starting in exactly 12 hours (+/- 15 min window) and 1 hour (+/- 15 min window)
For each session:
  Get all confirmed bookings for session
  For each booked member:
    Idempotency check: class-reminder-12h or class-reminder-1h already sent for this session+member?
    If not sent:
      Call send-message.json with:
        Template: class-reminder
        Variables: member_name, class_name, date, time, trainer_name, location
      Record in Notification Log (event_id: session_id + member_id + reminder_type)
```

TEMPLATES USED	class-reminder
VARIABLES	member_name, class_name, date, time, trainer_name, location
TIMING	12 hours before session; 1 hour before session (configurable)

8.5 Workflow 04: Membership Renewal Reminders

FILE	renewal-reminders.json
TRIGGER	n8n scheduled job (runs daily at configured time, e.g., 9:00 AM)
PURPOSE	Remind members to renew their membership before it expires.

FLOW

```
Daily scheduled trigger
Call API: getExpiringMemberships(days=[7, 3, 1, 0])
  API returns: members with memberships expiring in exactly 7 days, 3 days, 1 day, and today (0)
For each group:
  Select correct template:
    7 days: renewal-reminder-7d
    3 days: renewal-reminder-3d
    1 day: renewal-reminder-1d
    0 days (expiry day): membership-expired
  For each member in group:
    Idempotency check: same reminder type already sent for this membership_id?
    If not sent:
      Call send-message.json with template and variables
      Record in Notification Log (event_id: membership_id + reminder_type)
```

TEMPLATES USED	renewal-reminder-7d, renewal-reminder-3d, renewal-reminder-1d, membership-expired
VARIABLES	member_name, plan_name, expiry_date, renewal_link
TIMING	Daily scheduled; membership expiry determines which template fires

8.6 Workflow 05: Payment Received

FILE	payment-received.json
TRIGGER	Outbox event — payment.confirmed
PURPOSE	Send immediate payment confirmation to member.

FLOW

```
Receive signed webhook (event_type: payment.confirmed)
Verify HMAC signature
Idempotency check (event_id)
Extract: member_id, amount, purpose, payment_reference from payload
Fetch member phone from Members service
Call send-message.json with:
  Template: payment-received
  Variables: member_name, amount, payment_reference, purpose, date
Record in Notification Log
```

TEMPLATE	payment-received
VARIABLES	member_name, amount, payment_reference, purpose, date
TIMING	Immediately after payment.confirmed outbox event

8.7 Workflow 06: Event Registration

FILE	event-registration.json
TRIGGER	Outbox event — event.registration.confirmed
PURPOSE	Confirm event registration to registrant immediately.

FLOW

```
Receive signed webhook (event_type: event.registration.confirmed)
Verify HMAC signature
Idempotency check (event_id)
Extract: member_id, event_id, registration_id from payload
Fetch event details: name, date, time, location, price
Fetch member phone from Members service
Call send-message.json with:
  Template: event-registration-confirmation
  Variables: member_name, event_name, date, time, location, amount_paid_or_free
Record in Notification Log
```

TEMPLATE	event-registration-confirmation
TIMING	Immediately after registration confirmed

8.8 Workflow 07: Event Reminder

FILE	event-reminders.json
TRIGGER	n8n scheduled job (runs every 30 minutes)
PURPOSE	Send reminders at 24 hours, 6 hours, and 1 hour before each event.

FLOW

```
Scheduled trigger fires (every 30 minutes)
Call API: getEventsWithRegistrations(starts_in_hours=[24, 6, 1])
For each event:
  Get all confirmed registrations
  For each registrant:
    Idempotency check: reminder for this event+registrant+reminder_type already sent?
    If not sent:
      Call send-message.json with:
        Template: event-reminder
        Variables: member_name, event_name, date, time, location
      Record in Notification Log
```

TEMPLATES	event-reminder
VARIABLES	member_name, event_name, date, time, location
TIMING	24h, 6h, 1h before event start (configurable in Settings)

8.9 Workflow 08: Notification Send Sub-Workflow (Shared)

FILE	send-message.json
TYPE	Called sub-workflow (not triggered by outbox event directly)
PURPOSE	Shared message dispatch logic with mock/live toggle, retry, and status recording.

INPUTS (FROM CALLING WORKFLOW)

Parameter	Type	Description
phone	string	Recipient's WhatsApp phone number (with country code)
template_name	string	Approved WhatsApp template identifier
variables	object	Template variable values
event_id	string	Idempotency key from calling workflow
tenant_id	string	Tenant context for Notification Log scoping
member_id	string or null	Member reference for Notification Log

FLOW

```
Receive inputs from calling workflow
Check MOCK_MODE environment variable:

MOCK MODE (Phase 1):
  Write Notification Log record:
    recipient: phone
    template: template_name
    variables: variables (as JSON)
    status: mock
    event_id: event_id
```

```
    timestamp: now()
  Return: {status: "mock", logged: true}
  No API call to WhatsApp

LIVE MODE (Phase 2+):
  Call WhatsApp Cloud API:
    POST /messages
    to: phone
    type: template
    template: {name: template_name, components: [variables]}
  Handle response:
    Success → status: sent
    Error → status: failed, error: error_code
  Write Notification Log record with status
  If failed: trigger failure-alert.json for this specific delivery
  Return: {status: "sent|failed", whatsapp_message_id: "..."}
```

RETRY POLICY	3 attempts with exponential backoff for WhatsApp API calls in Live mode only.

8.10 Workflow 09: Failure Alert

FILE	failure-alert.json
TRIGGER	Called by send-message.json after all retries exhausted; OR called by any workflow on critical error.
PURPOSE	Surface delivery failures to Admin notification panel.

FLOW

```
Receive failure context:
  workflow_name, event_id, recipient, template, error_code, tenant_id

Create Notification Log record with status: failed
Call API: createAdminAlert (type: notification_failure, payload: failure context)
  API creates admin alert visible in B2B notification panel

Optional (configurable): notify Platform Admin if failure rate exceeds threshold
```

8.11 Automation Testing and Mock Mode Spec

Mock Mode Behaviour (Phase 1)

In mock mode, every outbox event that would trigger a WhatsApp message instead:

1. Writes a Notification Log record with status: mock
2. Stores all template variables and recipient as if the message were sent
3. Does NOT call the WhatsApp Cloud API
4. Returns success to the calling workflow

Admin Mock Mode Inspection

Admin can navigate to B2B Notifications module and:

- View all mock-mode would-send records
- See: recipient phone, template name, variables, timestamp
- Verify that correct templates fire for each event
- Verify idempotency: same event only triggers one record per recipient

Production Cutover (Live Mode Activation)

To switch from mock to live:

1. Owner configures production WhatsApp Business Account credentials in Settings
2. Platform Admin sets MOCK_MODE=false environment variable in n8n
3. No workflow code changes required
4. All subsequent events processed in live mode

Testing Requirements (QA Team)

- QA must verify each workflow fires for its trigger event (mock mode inspection)
- QA must verify idempotency: trigger same event twice → only one Notification Log record per recipient
- QA must verify HMAC signature rejection: send unsigned webhook → workflow rejects
- QA must verify all template variables populated correctly
- QA must verify failure-alert workflow fires when send-message fails after all retries

SECTION 9

Module Dependencies and Integration Contracts

9.1 Dependency Map (Read as: Module → Depends On)

B2B Module Dependencies

Module	Depends On	Dependency Type
Dashboard	Members, Memberships, Bookings, Attendance, Payments, Reports	Data (aggregated reads)
CRM	Members (on conversion), Notifications, Settings (source list)	Data + Write (conversion)
Members	Memberships (status derivation), Bookings, Attendance, Payments	Data
Memberships	Members, Payments, Settings (plan definitions), Notifications	Data + Write
Classes	Staff Management (trainer reference), Settings (class types), Bookings	Data + Configuration
Bookings	Members, Memberships (eligibility), Classes (capacity), Payments, Notifications	Data + Transactional
Attendance	Bookings (booking validation), Members, Classes (session), Staff Management	Data + Transactional
Events	Members, Payments, Notifications, Settings (event categories)	Data + Transactional
Payments	Members, Memberships (activation), Events (registration), Gateway (Razorpay)	Data + External
Reports	All modules (read aggregates via reporting service)	Data (read-only)
Staff Management	Auth (JWT), Settings (role config)	Configuration
Settings	All configurable modules (provides configuration to all)	Configuration
Notifications	Outbox (event source), n8n (orchestration), WhatsApp Cloud API	External + Automation
Tenant Management	All modules (provides tenant context to all)	Cross-cutting
AI Assistant B2B	Members, Memberships, Attendance, Reports, AI Platform	Data + AI

B2C Module Dependencies

Module	Depends On	Dependency Type
B2C Dashboard	Memberships, Bookings, Attendance	Data
B2C Membership	Settings (active plans), Payments (checkout)	Data + Transactional
B2C Bookings	Classes (schedule), Memberships (eligibility), Payments	Data + Transactional
B2C Attendance	Bookings (booking validation), Classes (session)	Data + Transactional
B2C Events	Events (listings), Payments (checkout)	Data + Transactional
B2C Payments	Memberships, Events, Gateway (Razorpay)	Transactional + External
B2C Profile	Auth (OTP, Google OAuth), Members	Data + External
B2C Notifications	n8n, WhatsApp Cloud API	External + Automation
B2C Recommendations	Members (profile), Attendance (summary), Classes (schedule), AI Platform	Data + AI

9.2 ERP Integration Contract

Critical Rule (Restated)

All business logic that must be atomic, transactional, or authoritative executes inside ERPNext (custom Frappe app). The API layer orchestrates and routes; it does not duplicate business rules that live in ERP.

ERP Custom App Whitelisted Methods

These methods are exposed via ERPNext's `frappe.call` whitelist or REST API and called by the NestJS API:

Method	Belongs To	Called For
`wellness_studio.api.booking.create_booking`	Booking DocType	Atomic seat booking with eligibility, capacity, and duplicate checks
`wellness_studio.api.booking.cancel_booking`	Booking DocType	Booking cancellation and waitlist promotion trigger
`wellness_studio.api.booking.promote_waitlist`	Booking DocType	Waitlist promotion (called after cancellation)
`wellness_studio.api.attendance.check_in`	Attendance Record DocType	QR check-in with duplicate and booking validation
`wellness_studio.api.attendance.mark_attendance`	Attendance Record DocType	Trainer bulk attendance marking
`wellness_studio.api.membership.activate_membership`	Membership DocType	State transition Pending to Active

Method	Belongs To	Called For
`wellness_studio.api.membership.expire_membership`	Membership DocType	State transition to Expired (scheduler-triggered)
`wellness_studio.api.membership.suspend_membership`	Membership DocType	State transition to Suspended
`wellness_studio.api.member.create_member`	Member DocType	Member creation with duplicate check
`wellness_studio.api.payment.record_payment`	Payment Record DocType	Manual payment recording
`wellness_studio.api.payment.confirm_payment`	Payment Record DocType	Gateway webhook → payment status update
`wellness_studio.api.event.register`	Event Registration DocType	Event registration with capacity check
`wellness_studio.api.lead.create_lead`	Lead DocType	Lead creation with duplicate check
`wellness_studio.api.lead.convert_lead`	Lead + Member DocType	Lead conversion with source preservation

ERP to API Callbacks (ERP → API)

The ERP custom app uses doc_events and scheduled jobs to call back to the API:

Trigger	ERP Hook	API Endpoint Notified
Membership expiry date reached	`Frappe scheduler`	`POST /internal/events/membership.expiring`
Membership expires	`Frappe scheduler`	`POST /internal/events/membership.expired`
Class session starts	`Frappe scheduler`	`POST /internal/events/session.starting`
Booking count reaches capacity	`doc_event on Booking`	`POST /internal/events/session.full`

All callbacks are signed with internal HMAC secret and validated by NestJS internal event intake.

ERP Tenant Scoping Implementation

Every ERP DocType in the custom app includes a `tenant_id` field. ERPNext permission hooks enforce:

```
# hooks.py
permission_query_conditions = {
    "Wellness Member": "wellness_studio.permissions.get_tenant_condition",
    "Wellness Membership": "wellness_studio.permissions.get_tenant_condition",
    # ... all DocTypes
}
```

The `get_tenant_condition` function returns a SQL WHERE clause filtering by the authenticated user's tenant_id. This is enforced at the ORM layer — no raw SQL bypasses it.

9.3 Payment Gateway Integration Contract

Provider: Razorpay

PHASE 1	Test mode credentials. No real money processed.
PHASE 2	Production credentials; switch by environment variable only.

ORDER CREATION (API → RAZORPAY)

```
POST https://api.razorpay.com/v1/orders
Auth: Basic base64(key_id:key_secret)
Body: {
  amount: <paise integer — INR * 100>,
  currency: "INR",
  receipt: "<internal payment ID>",
  notes: { purpose: "membership|event", tenant_id: "<id>", member_id: "<id>" }
}
Response: { id: "order_...", amount: ..., currency: "INR", status: "created" }
```

PAYMENT VERIFICATION (RAZORPAY → API WEBHOOK)

```
POST /api/webhooks/razorpay
Headers: X-Razorpay-Signature: <HMAC-SHA256 of payload using webhook_secret>
Body: { event: "payment.captured|payment.failed", payload: { payment: { entity: {...} } } }

Verification:
  generated_signature = HMAC-SHA256(webhook_body, RAZORPAY_WEBHOOK_SECRET)
  if generated_signature != X-Razorpay-Signature: return 401 and discard
  else: process event atomically
```

IDEMPOTENCY	Same `razorpay_payment_id` never processed twice; checked before status update.

9.4 WhatsApp Cloud API Integration Contract

PROVIDER	Meta WhatsApp Business Cloud API
PHASE 1	Mock mode — API not called; Notification Log records written
PHASE 2	Live mode — production credentials from Settings

SEND MESSAGE (N8N → WHATSAPP API)

```
POST https://graph.facebook.com/v18.0/{phone_number_id}/messages
Headers: Authorization: Bearer {access_token}
Body: {
  messaging_product: "whatsapp",
  to: "<recipient phone with country code e.g. 919876543210>",
  type: "template",
  template: {
    name: "<template_name>",
    language: { code: "en" },
    components: [
      { type: "body", parameters: [ { type: "text", text: "<variable_value>" }, ... ] }
    ]
  }
}
Response: { messages: [{ id: "wamid...." }] }
```

DELIVERY STATUS WEBHOOK (META → N8N)

```
POST /webhooks/whatsapp/status
Body: { entry: [{ changes: [{ value: { statuses: [{ id: "wamid...", status: "sent|delivered|read|
failed" }] } }] }] }
n8n updates Notification Log record with delivery status for corresponding wamid
```

9.5 Auth Integration Contract

Phone OTP (B2C Member Login)

```
Flow:
  Member enters phone → POST /auth/otp/send { phone, tenantId }
  API validates phone format → OTP generated (6 digits) → Stored in Redis/DB with TTL=600s
  OTP sent to phone via WhatsApp or SMS (provider configurable)

  Member enters OTP → POST /auth/otp/verify { phone, otp, tenantId }
  API validates OTP: correct + not expired + not already used
  If valid: OTP marked used → JWT issued { memberId, tenantId, role: "member" }
  JWT TTL: 7 days (configurable); refresh token: 30 days
```

Google OAuth (B2C Member Login)

```
Flow:
  Member taps Login with Google → GET /auth/google?tenantId=<id>
  API redirects to Google OAuth consent screen

  Google redirects to: GET /auth/google/callback?code=<code>&state=<tenantId>
  API exchanges code for Google ID token
  API validates ID token with Google tokeninfo endpoint
  Extracts: google_user_id, email

  Looks up account link: SELECT member_id FROM google_accounts WHERE google_user_id = $1 AND tenant_id = $2
  If found: JWT issued for linked member
  If not found: Return { action: "link_required", email: <google_email> }
    → Member prompted to enter phone to link accounts
    → OTP verification links google_user_id to member_id
    → JWT issued
```

B2B Staff Login (Email + Password / SSO future)

```
Flow (Phase 1):
  Staff enters email + password → POST /auth/staff/login { email, password, tenantId }
  API validates credentials against Staff DocType (bcrypt password hash)
  If valid: JWT issued { staffId, tenantId, role: "owner|admin|front_desk|trainer" }
  JWT TTL: 8 hours (configurable for staff)

  MFA (Phase 2):
    After password validation, TOTP challenge sent to authenticator app
    Staff enters 6-digit TOTP → validated
    JWT issued only after TOTP valid
```

9.6 External Dependency Versions and Stability

Dependency	Phase 1 Version /Notes	Phase 2 Notes
ERPNext	v15.x (Frappe v15)	Pin major version; upgrade path documented
Frappe Framework	v15.x	Custom app compatibility tested against this version
NestJS	v10.x	API framework
PostgreSQL	v16.x	Application DB
Redis	v7.x	Session/cache
n8n	v1.x (self-hosted)	Workflow orchestration
Razorpay SDK	Latest stable at build time	Test mode in Phase 1
WhatsApp Cloud API	v18.0	Mock mode in Phase 1
LLM Provider	Gemini 1.5 Pro or GPT-4o (configured per environment)	Model-agnostic adapter; swap without code change

Dependency	Phase 1 Version /Notes	Phase 2 Notes
Google OAuth	OAuth 2.0 standard	Stable

SECTION 10

MVP Implementation Scope (5-Day Delivery)

10.1 Scope Statement

The 5-day MVP is a fully functional, production-deployable subset of the platform that demonstrates the core value proposition: a single studio can onboard members, manage class schedules, take bookings, record attendance, run events, track payments, and communicate with members via WhatsApp — all through a unified, governed, multi-tenant platform.

> **The MVP is NOT a prototype.** It is production-quality code running on the full architecture. Phase 2 modules are added to this foundation — not rebuilt.

10.2 P0 — 5-Day MVP Scope (All modules marked P0)

B2B Portal — P0 Screens and Functions

AUTHENTICATION

- B2B-A-01 Staff Login (email + password)
- B2B-A-02 Forgot Password
- B2B-A-04 Accept Staff Invitation
- JWT issuance and validation middleware
- Role-based route guards for all B2B routes

DASHBOARD (B2B-D-01)

- Active members count
- Sessions today count
- Today's class schedule with occupancy
- Renewal pipeline (expiring in 7 days)
- Quick Actions: Add Member, Capture Lead, Check In
- Revenue tiles (Owner/Admin only)

MEMBERS — FULL CRUD

- B2B-M-01 Member List with search and filter
- B2B-M-02 to M-08 Member Detail, Create, Edit forms
- Duplicate detection modal

- Memberships, Attendance, Payments tabs on Member Detail

MEMBERSHIPS

- B2B-MS-01 Plan List; B2B-MS-02 Plan Create/Edit
- B2B-MS-03 Assign Membership
- B2B-MS-04 Approval Queue
- B2B-MS-05 Renewal Pipeline view
- State machine: Pending → Active → Expired → Suspended → Cancelled

CLASSES

- B2B-CL-01 Weekly Schedule view
- B2B-CL-02 Session Detail with participant list
- B2B-CL-03 Create Session (single and recurring)
- B2B-CL-04 Edit Session
- B2B-CL-05 Class Type Management
- B2B-CL-07 Cancel Session Modal

BOOKINGS

- B2B-BK-02 Book on Behalf Modal (Admin/FD)
- B2B-BK-03 Cancel Booking Modal
- Session bookings list in Session Detail
- Waitlist view per session

ATTENDANCE

- B2B-AT-01 Check-In Dashboard (front desk)
- B2B-AT-02 Session Attendance Sheet (marking)
- B2B-AT-04 QR Code Display Screen
- B2B-AT-05 Attendance Correction Modal (with audit)

EVENTS

- B2B-EV-01 through EV-07 full Events module
- Paid and free events
- Registration management

PAYMENTS

- B2B-PY-01 Payment List
- B2B-PY-02 Record Manual Payment Modal
- Razorpay integration (test mode)
- Gateway callback handler with signature verification

STAFF MANAGEMENT

- B2B-ST-01 Staff List
- B2B-ST-03 Invite Staff

- B2B-ST-04 Role Assignment
- B2B-ST-06 Deactivate Staff Modal

SETTINGS (CONFIGURATION CENTER)

- B2B-SE-01 Settings Hub
- B2B-SE-02 Studio Profile
- B2B-SE-03 Plan Management
- B2B-SE-04 Class Type Management
- B2B-SE-06 Feature Flags
- B2B-SE-07 Booking Rules (cancellation window)

TENANT MANAGEMENT

- Single pilot studio onboarding
- Multi-tenant model active from day one (no special-casing for single tenant)
- Tenant, Studio, Branch hierarchy in ERP

NOTIFICATIONS (INFRASTRUCTURE)

- Outbox event emission for all P0 events
- n8n workflows: all 8 workflows deployed in mock mode
- Notification Log records written for all events
- Admin Notification Log view (B2B-NT-01)

ERP CUSTOM APP

- All 19 DocTypes deployed
- All whitelisted API methods for P0 operations
- Tenant scoping hooks active
- Audit Log for all transactional operations
- ERP expiry scheduler for membership renewal pipeline

B2C Portal — P0 Screens and Functions

AUTHENTICATION (B2C)

- B2C-A-01 Phone OTP Login
- B2C-A-02 OTP Verification
- B2C-A-03 Google OAuth Callback
- B2C-A-04 Account Link Flow
- B2C-A-05 New Member Registration

DASHBOARD (B2C-D-01)

- Membership status card with plan, validity, status badge
- Upcoming bookings (next 3)
- Renewal CTA when expiring ≤7 days or Expired

MEMBERSHIP

- B2C-MS-01 Plan Browsing
- B2C-MS-03 Purchase Checkout (Razorpay test mode)
- B2C-MS-04 Payment Confirmation
- B2C-MS-05 Membership History

BOOKINGS

- B2C-BK-01 Class Schedule with eligibility filtering
- B2C-BK-02 through BK-07 full self-booking flow
- Cancellation within window
- Waitlist join

ATTENDANCE

- B2C-AT-01 QR Scanner / Check-In
- B2C-AT-02 Check-In Success Screen
- B2C-AT-03 My Attendance History

EVENTS

- B2C-EV-01 through EV-04 Event discovery and registration

PROFILE

- B2C-PR-01 My Profile view
- B2C-PR-02 Edit Profile

PAYMENTS

- B2C-PY-01 Payment History

10.3 P1 — Phase 1 Additions (Post-MVP; Within First Sprint)

Module	Additions
Reports	Full reports hub (6 core reports)
AI Admin Assistant	2 read queries + Create Member action
AI Enquiry Assistant	Public enquiry and lead capture
AI Booking Assistant	Eligible session shortlist
AI Recommendation Assistant	Goal/level/history-based suggestions on B2C dashboard
AI Feedback Assistant	Post-class comment classification
n8n Live Mode	Production WhatsApp Cloud API activation
Staff Management	Granular permission grants

Module	Additions
Notifications (B2B)	Full Notification Log with delivery status
Settings	Notification template management

10.4 P2 — Phase 2 (Second Sprint / Future)

Feature	Module
MFA for Owner and Admin	Auth / Staff Management
Recurring payment / auto-renewal	Memberships / Payments
In-app notification centre	Notifications
CRM Agent (lead scoring, draft follow-up)	AI Agents
Operations Agent (capacity monitoring)	AI Agents
Report export CSV/PDF	Reports
Revenue reconciliation	Payments
Multi-studio dashboard	Dashboard
No-show processing automation	Attendance / n8n
Inactive member nudge workflow	n8n
Refunds through gateway	Payments
Member deactivation self-service	Profile
GST invoicing	Payments / ERP

10.5 Acceptance Criteria Summary (MVP Gate)

The MVP is production-ready and deployable when all of the following are verified:

Functional Gate

#	Criterion	Verified By
F-01	Member creation completes in under 1 minute end-to-end	QA — operational test
F-02	Class booking (member self-serve): 95th percentile under 1 second	Load test
F-03	Last-seat contention: exactly one success for N concurrent requests	Concurrent test (NFR-PRF-03)
F-04	Membership expiry blocks class booking at server level	QA — boundary test
F-05	Payment marked Paid only after server-side Razorpay signature verified	QA — webhook test

#	Criterion	Verified By
F-06	Cross-tenant data access denied at server level (returns 403)	QA — tenant isolation test
F-07	All mock-mode notifications write correct Notification Log records	QA — mock mode inspection
F-08	Attendance correction creates immutable audit record	QA — audit log test
F-09	Role-scoped views enforce permissions at API level (not just UI)	QA — API penetration test
F-10	Duplicate member detection: same phone within tenant shows modal before save	QA — duplicate test

Performance Gate (NFR)

#	Criterion	Target
NFR-PRF-01	API response time (95th percentile)	< 500ms under normal load
NFR-PRF-02	Page load time (initial)	< 3 seconds
NFR-PRF-03	Last-seat concurrent booking	Exactly 1 success

Security Gate

#	Criterion
SEC-01	JWT validation required on every protected route
SEC-02	Tenant ID in JWT matched against all resource queries
SEC-03	Razorpay webhook signature verification mandatory
SEC-04	n8n webhook HMAC verification mandatory
SEC-05	No sensitive data (passwords, API keys) in logs or error responses
SEC-06	All staff-modifiable fields produce audit records

Data Integrity Gate

#	Criterion
DI-01	No orphaned Booking records if capacity check fails
DI-02	No duplicate Attendance records per member per session
DI-03	No duplicate Payment records for same Razorpay payment ID
DI-04	Member phone unique within tenant — enforced at ERP level
DI-05	All state transitions produce audit log entries

10.6 Non-Functional Requirements Summary

Performance (NFR-PRF)

ID	Requirement	Target
NFR-PRF-01	API response time 95th percentile	< 500ms under normal load
NFR-PRF-02	Page load time	< 3 seconds
NFR-PRF-03	Concurrent booking last seat	Exactly 1 success
NFR-PRF-04	Notification delivery (live mode)	< 2 minutes from trigger event

Scalability (NFR-SCL)

ID	Requirement
NFR-SCL-01	Multi-tenant data model active from day one; single pilot studio runs on same model without special-casing
NFR-SCL-02	Adding a new studio requires configuration only — no code change
NFR-SCL-03	New wellness vertical (class type) requires configuration only — no hard-coded type names

Reliability (NFR-REL)

ID	Requirement
NFR-REL-01	Booking and payment operations are atomic; partial states not possible
NFR-REL-02	Outbox pattern ensures notifications sent eventually even if n8n temporarily down
NFR-REL-03	ERP and API DB failures do not corrupt each other's data

Security (NFR-SEC)

ID	Requirement
NFR-SEC-01	All routes protected by JWT; no unauthenticated access to business data
NFR-SEC-02	Tenant isolation enforced at ORM layer in ERP and query layer in API
NFR-SEC-03	Secrets (API keys, credentials) never logged, never returned in API responses
NFR-SEC-04	Audit log is append-only; no update or delete operations permitted
NFR-SEC-05	AI Audit log is append-only; no update or delete operations permitted

Observability (NFR-OBS)

ID	Requirement
NFR-OBS-01	Structured logging on all API requests with request ID, tenant ID, user ID
NFR-OBS-02	All ERP transactions logged with document ID, actor, before/after
NFR-OBS-03	n8n workflow execution history retained for at least 30 days
NFR-OBS-04	AI Audit Log queryable by tenant, user, tool, and date range

10.7 Technology Stack Summary (Implementation Reference)

Backend (API)

- **Framework:** NestJS (TypeScript) — modular monolith
- **Database:** PostgreSQL (identity, config, audit, notifications, AI audit, outbox)
- **Cache:** Redis (session, OTP, rate limiting)
- **ERP:** ERPNext v15 / Frappe v15 (custom app: wellness_studio)
- **Auth:** JWT (access + refresh); bcrypt for passwords; Google OAuth 2.0
- **Payment:** Razorpay SDK (test mode Phase 1)
- **Queue/Outbox:** PostgreSQL-backed outbox pattern; polling dispatcher

Frontend (B2B Portal)

- **Framework:** Next.js (React) — server-side rendering for SEO and performance
- **UI Library:** To be confirmed (shadcn/ui or custom component library)
- **State:** React Query (server state); Zustand (client state)
- **Auth:** JWT stored in httpOnly cookies

Frontend (B2C Member Portal)

- **Framework:** Next.js (React) — PWA capability for mobile check-in
- **UI:** Mobile-first responsive design
- **QR Scanner:** Browser camera API / react-qr-reader
- **Auth:** JWT in httpOnly cookies; Google OAuth redirect flow

Automation

- **Engine:** n8n (self-hosted, v1.x)
- **Messaging:** WhatsApp Cloud API (Meta) — mock mode Phase 1
- **Trigger:** Outbox events via HMAC-signed webhooks

AI Platform

- **LLM Gateway:** Model-agnostic NestJS adapter
- **Phase 1 Model:** Google Gemini 1.5 Pro (primary) or GPT-4o (alternate)
- **Orchestration:** NestJS AI Orchestrator service
- **Audit:** PostgreSQL append-only AI Audit table
- **Evaluation:** Jest test harness with curated query set

Infrastructure

- **Hosting:** Cloud VMs or container orchestration (to be specified by DevOps)
- **Secrets Management:** Environment variables + secrets manager (to be specified)
- **TLS:** HTTPS enforced on all endpoints; no HTTP in production
- **Monitoring:** Structured logs to log aggregator; alerting on error rate thresholds

10.8 Implementation Notes for Development Teams

For Frontend Developers

- Every screen in the Screen Inventory (Section 4) has a defined route and role access
- All data fetching is via API — no direct database or ERP queries from frontend
- Role-gated UI elements must still be enforced at API level; UI hiding is UX only
- Loading states: every data-dependent component shows skeleton → populated → error states
- Error handling: API errors shown in user-friendly language; raw error codes not exposed to users
- Optimistic UI: never use optimistic updates for bookings or payments — always wait for authoritative API response

For Backend Developers

- Auth middleware validates JWT on every route; tenant_id extracted from JWT, not from URL params
- All ERP calls go through the ERP client service — never raw HTTP to ERP from feature modules
- Outbox pattern: every business event appended to outbox table atomically with the transaction that produced it
- Idempotency: all POST endpoints that create business records check for existing records with same natural key before creating new
- Audit: all write operations (create/update/state change) produce audit log entry in same transaction

For ERP / Frappe Developers

- Custom app namespace: wellness_studio — all DocTypes, hooks, and API methods prefixed
- Permission hooks active on all DocTypes — never skip get_tenant_condition
- State machine transitions enforced in custom validation hooks — invalid transitions throw exceptions
- Whitelisted API methods are the only ERP entry points used by the NestJS API

- ERP scheduler jobs for expiry and reminders must be idempotent — running twice for same period produces no duplicates

For AI Developers

- Tool definitions are the contract — every tool has: name, description, input schema, output schema
- LLM calls are made via the AI Gateway adapter — never call LLM provider SDK directly from feature modules
- Pending Action store: arguments stored at extraction time; retrieved at confirmation time; never re-extracted
- AI Audit record creation is mandatory — a missing audit record for any AI call is a bug
- Evaluation harness must run on every CI/CD pipeline run; test result reported alongside build status

For QA Team

- Test strategy must include: functional tests for each module, boundary tests for business rules, role-scope tests (API level), tenant isolation tests, concurrent booking test, payment failure tests
- Mock mode notification testing: trigger each business event and verify Notification Log record
- AI accuracy testing: run curated query set against AI Admin Assistant; confirm 85%/95%/zero targets met
- Performance testing: booking flow under concurrent load; page load time measurement
- Data integrity: verify no orphaned records after failed transactions

APPENDIX A

Document Index and Coverage

File	Section	Contents
DOC3_SEC1_Module_Inventory.md	1	Complete module inventory — B2B, B2C, AI, Automation, ERP
DOC3_SEC2_B2B_Specifications.md	2	Full B2B module specifications (15 modules)
DOC3_SEC3_B2C_Specifications.md	3	Full B2C module specifications (9 modules)
DOC3_SEC4_SEC5_Screens_Workflows.md	4 and 5	Screen inventory (all screens) and cross-module workflows
DOC3_SEC6_SEC7_SEC8_AI_Automation.md	6, 7, 8	AI module specs, AI Agent specs, Automation workflow specs
DOC3_SEC9_SEC10_Dependencies_MVP.md	9 and 10	Module dependencies, integration contracts, MVP scope

TOTAL COVERAGE

- B2B Modules: 15 fully specified
- B2C Modules: 9 fully specified
- AI Modules Phase 1: 5 fully specified
- AI Agents Phase 2+: 5 outlined
- Automation Workflows: 9 fully specified
- B2B Screens: 67 identified
- B2C Screens: 28 identified
- Public Screens: 6 identified
- Cross-Module Workflows: 8 fully documented
- MVP Acceptance Criteria: 20 functional, 3 performance, 6 security, 5 data integrity
- ERP DocTypes: 19 specified
- Integration Contracts: Razorpay, WhatsApp, Google OAuth, ERP API documented